# NEET

Time : 3 H 20 min
MM : 720

# Solved Paper 2022

**Instructions**

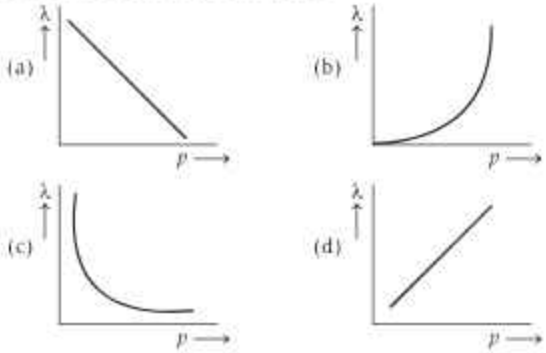
- This question paper contains 200 Multiple Choice Question from the subjects **Physics, Chemistry** & **Biology** (Botany & Zoology). There are 50 questions in each subject, divided into two Sections : Section-A and Section B.
- **Section A** contains 35 questions in each subject (Q. Nos. 1 to 35, 51 to 85, 101 to 135 and 151 to 185 ) and all questions are compulsory.
- **Section B** contains 15 questions from each subject (Q. Nos. 36 to 50, 86 to 100, 136 to 150 and 186 to 200) out of which 10 questions are to be attempted.
- Each question carries 4 Marks. For each correct response, the candidate will get 4 marks. For each incorrect response 1 mark will be deducted from the total scores.

## PHYSICS

### Section-A

**1.** If a soap bubble expands, the pressure inside the bubble
(a) increases
(b) remains the same
(c) is equal to the atmospheric pressure
(d) decreases.

**2.** The graph which shows the variation of the de-Broglie wavelength ($\lambda$) of a particle and its associated momentum ($p$) is




**3.** A body of mass 60 g experiences a gravitational force of 3.0 N, when placed at a particular point. The magnitude of the gravitational field intensity at that point is
(a) 50 N/kg (b) 20 N/kg (c) 180 N/kg (d) 0.05 N/kg

**4.** Given below are two statements

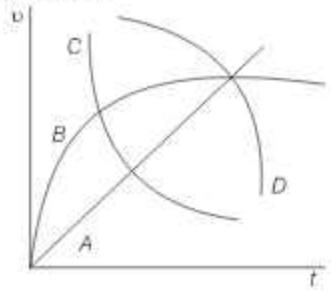
**Statement I** Biot-Savart's law gives us the expression for the magnetic field strength of an infinitesimal current element ($Idl$) of a current carrying conductor only.

**Statement II** Biot-Savart's law is analogous to Coulomb's inverse square law of charge $q$, with the former being related to the field produced by a scalar source, $Idl$ while the latter being produced by a vector source, $q$. In light of above statements choose the most appropriate answer from the options given below
(a) Both Statement I and Statement II are incorrect.
(b) Statement I is correct and Statement II is incorrect.
(c) Statement I is incorrect and Statement II is correct.
(d) Both Statement I and Statement II are correct.

**5.** The ratio of the distances travelled by a freely falling body in the 1st, 2nd, 3rd and 4th second
(a) 1 : 4 : 9 : 16 (b) 1 : 3 : 5 : 7
(c) 1 : 1 : 1 : 1 (d) 1 : 2 : 3 : 4

**6.** In half wave rectification, if the input frequency is 60 Hz, then the output frequency would be
(a) 30 Hz (b) 60 Hz (c) 120 Hz (d) zero

**7.** A spherical ball is dropped in a long column of a highly viscous liquid. The curve in the graph shown, which represents the speed of the ball ($v$) as a function of time ($t$) is




(a) $B$ (b) $C$ (c) $D$ (d) $A$

***Note*** *(*) Marked questions are from the topics not included in latest NEET syllabus.*

**8.** The angle between the electric lines of force and the equipotential surface is
(a) 45° (b) 90° (c) 180° (d) 0°

**9.** As the temperature increases, the electrical resistance
(a) decreases for both conductors and semiconductors.
(b) increases for conductors but decreases for semiconductors.
(c) decreases for conductors but increases for semiconductors.
(d) increases for both conductors and semiconductors.

**10.** In the given nuclear reaction, the element $X$ is

$$^{22}_{11}\text{Na} \longrightarrow X + e^{+} + \nu$$

(a) $^{23}_{10}$Ne (b) $^{22}_{10}$Ne
(c) $^{22}_{12}$Mg (d) $^{23}_{11}$Na

**11.** Let $T_1$ and $T_2$ be the energy of an electron in the first and second excited states of hydrogen atom, respectively. According to the Bohr's model of an atom, the ratio of $T_1 : T_2$ is:
(a) 4 : 1 (b) 4 : 9
(c) 9 : 4 (d) 1 : 4

**12.** The ratio of the radius of gyration of a thin uniform disc about an axis passing through its centre and normal to its plane to the radius of gyration of the disc about its diameter is
(a) $\sqrt{2}$ :1 (b) 4 : 1 (c) 1 : $\sqrt{2}$ (d) 2 : 1

**13.** When light propagates through a material medium of relative permittivity $\epsilon_r$ and relative permeability $\mu_r$, the velocity of light, $v$ is given by ($c$-velocity of light in vacuum)
(a) $v = \sqrt{\frac{\mu_r}{\epsilon_r}}$ (b) $v = \sqrt{\frac{\epsilon_r}{\mu_r}}$
(c) $v = \frac{c}{\sqrt{\epsilon_r \mu_r}}$ (d) $v = c$

**14.** The energy that will be ideally radiated by a 100 kW transmitter in 1 hour is :
(a) $36 \times 10^4$ J (b) $36 \times 10^5$ J
(c) $1 \times 10^5$ J (d) $36 \times 10^7$ J

**15.** An electric lift with a maximum load of 2000 kg (lift + passengers) is moving up with a constant speed of 1.5 $ms^{-1}$. The frictional force opposing the motion is 3000 N. The minimum power delivered by the motor to the lift in watts is ($g = 10\ ms^{-2}$)
(a) 20000 (b) 34500
(c) 23500 (d) 23000

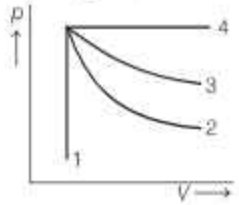
**16.** Two resistors of resistance, 100 Ω and 200 Ω are connected in parallel in an electrical circuit, The ratio of the thermal energy developed in 100 Ω to that in 200 Ω in a given time is
(a) 2 : 1 (b) 1 : 4
(c) 4 : 1 (d) 1 : 2

**17.** The peak voltage of the AC source is equal to
(a) the rms value of the AC source
(b) $\sqrt{2}$ times the rms value of the AC source
(c) $1/\sqrt{2}$ times the rms value of the AC source.
(d) the value of voltage supplied to the circuit.

**18.** A shell of mass $m$ is at rest initially. It explodes into three fragments having mass in the ratio 2 : 2 : 1. If the fragments having equal mass fly off along mutually perpendicular directions with speed $v$, the speed of the third (lighter) fragment is
(a) $\sqrt{2}v$ (b) $2\sqrt{2}v$
(c) $3\sqrt{2}v$ (d) $v$

**19.** If the initial tension on a stretched string is doubled, then the ratio of the initial and final speeds of a transverse wave along the string is
(a) $\sqrt{2}$ : 1 (b) 1 : $\sqrt{2}$
(c) 1 : 2 (d) 1 : 1

**20.** An ideal gas undergoes four different processes from the same initial state as shown in the figure below. Those processes are adiabatic, isothermal, isobaric and isochoric. The curve which represents the adiabatic process among 1, 2, 3 and 4 is

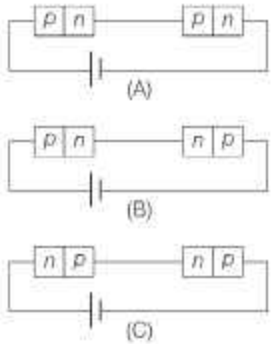



(a) 2 (b) 3 (c) 4 (d) 1

**21.** The angular speed of a fly wheel moving with uniform angular acceleration changes from 1200 rpm to 3120 rpm in 16 seconds. The angular acceleration in $rad/s^2$ is
(a) 4π (b) 12π (c) 104π (d) 2π

**22.** Plane angle and solid angle have
(a) Dimensions but no units
(b) No units and no dimensions
(c) Both units and dimensions
(d) Units but no dimensions

**23.**



In the given circuits (A), (B) and (C) the potential drop across the two *p-n* junctions are equal in
(a) Circuit (B) only (b) Circuit (C) only
(c) Both circuits (A) and (C) (d) Circuit (A) only

**24.** In a Young's double slit experiment, a student observes 8 fringes in a certain segment of screen when a monochromatic light of 600 nm wavelength is used. If the wavelength of light is changed to 400 nm, then the number of fringes he would observe in the same region of the screen is
(a) 8 (b) 9 (c) 12 (d) 6

**25.** When two monochromatic lights of frequency, ν and $\frac{\nu}{2}$ are incident on a photoelectric metal, their stopping potential becomes $\frac{V_s}{2}$ and $V_s$ respectively.
The threshold frequency for this metal is :
(a) 3ν (b) $\frac{2}{3}\nu$
(c) $\frac{3}{2}\nu$ (d) 2ν

**26.** A square loop of side 1 m and resistance 1 Ω placed in a magnetic field of 0.5 T. If the plane of loop is perpendicular to the direction of magnetic field, the magnetic flux through the loop is
(a) 0.5 Wb (b) 1 Wb
(c) zero Wb (d) 2 Wb

**27.** Two hollow conducting spheres of radii $R_1$ and $R_2 (R_1 > R_2)$ have equal charges. The potential would be
(a) more on smaller sphere
(b) equal on both the spheres
(c) dependent on the material property of the sphere
(d) more on bigger sphere

**28.** A Copper wire of length 10m and radius $(10^{-2}/\sqrt{\pi})$ m has electrical resistance of 10 Ω. The current density in the wire for an electric field strength of 10 (V / m) is
(a) $10^6$ A / m$^2$ (b) $10^{-5}$ A / m$^2$
(c) $10^5$ A / m$^2$ (d) $10^4$ A / m$^2$

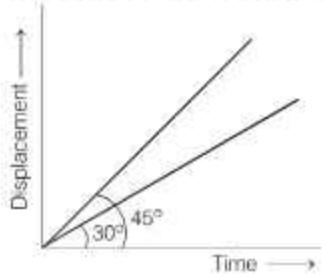
**29.** A long solenoid of radius 1 mm has 100 turns per mm. If 1A current flows in the solenoid, the magnetic field strength at the centre of the solenoid is
(a) $12.56 \times 10^{-2}$T (b) $12.56 \times 10^{-4}$T
(c) $6.28 \times 10^{-4}$T (d) $6.28 \times 10^{-2}$T

**30.** Two objects of mass 10 kg and 20 kg, respectively are connected to the two ends of a rigid rod of length 10 m with negligible mass. The distance of the center of mass of the system from the 10 kg mass is
(a) $\frac{20}{3}$ m (b) 10 m (c) 5 m (d) $\frac{10}{3}$ m

**31.** A light ray falls on a glass surface of refractive index $\sqrt{3}$ at an angle 60°. The angle between the refracted and reflected rays would be
(a) 60° (b) 90° (c) 120° (d) 30°

**32.** The dimensions $[MLT^{-2}A^{-2}]$ belong to the
(a) self inductance (b) magnetic permeability
(c) electric permittivity (d) magnetic flux

**33.** The displacement-time graph of two moving particles make angles of 30° and 45° with the *X*-axis as shown in the figure. The ratio of their respective velocity is




(a) 1 : 1 (b) 1 : 2
(c) 1 : $\sqrt{3}$ (d) $\sqrt{3}$ :1

**34.** A biconvex lens has radii of curvature, 20cm each. If the refractive index of the material of the lens is 1.5, the power of the lens is
(a) + 20 D (b) + 5 D (c) infinity (d) + 2 D

**35.** Match List - I with List - II

| | List-I (Electromagnetic waves) | List-II (Wavelength) |
|---|---|---|
| (A) | AM radio waves | (i) $10^{-10}$m |
| (B) | Microwaves | (ii) $10^{2}$m |
| (C) | Infrared radiations | (iii) $10^{-3}$m |
| (D) | X-rays | (iv) $10^{-4}$m |

Choose the correct answer from the options given below
(a) (A) - (iii), (B) - (ii), (C) - (i), (D) - (iv)
(b) (A) - (iii), (B) - (iv), (C) - (ii), (D) - (i)
(c) (A) - (ii), (B) - (iii), (C) - (iv), (D) - (i)
(d) (A) - (iv), (B) - (iii), (C) - (ii), (D) - (i)

## Section-B

**36.** A nucleus of mass number 189 splits into two nuclei having mass number 125 and 64. The ratio of radius of two daughter nuclei, respectively is:
(a) 4 : 5 (b) 5 : 4 (c) 25 : 16 (d) 1 : 1

**37.** Two transparent media *A* and *B* are separated by a plane boundary. The speed of light in those media are $1.5 \times 10^8$ m/s and $2.0 \times 10^8$ m / s, respectively. The critical angle for a ray of light for these two media is
(a) $\sin^{-1}(0.750)$ (b) $\tan^{-1}(0.500)$
(c) $\tan^{-1}(0.750)$ (d) $\sin^{-1}(0.500)$

**38.** A big circular coil of 1000 turns and average radius 10m is rotating about its horizontal diameter at 2 rad $s^{-1}$. If the vertical component of earth's magnetic field at that place is $2 \times 10^{-5}$T and electrical resistance of the coil is 12.56 Ω, then the maximum induced current in the coil will be

(a) 1.5 A (b) 1 A (c) 2 A (d) 0.25 A

**39.** Match List-I with List-II.

| | List-I | List-II |
|---|---|---|
| (A) | Gravitational constant ($G$) | (i) $[L^2T^{-2}]$ |
| (B) | Gravitational potential energy | (ii) $[M^{-1}L^3T^{-2}]$ |
| (C) | Gravitational potential | (iii) $[LT^{-2}]$ |
| (D) | Gravitational intensity | (iv) $[ML^2T^{-2}]$ |

Choose the correct answer from the options given below

(a) (A) - (ii), (B) - (iv), (C) - (i), (D) - (iii)
(b) (A) - (ii), (B) - (iv), (C) - (iii), (D) - (i)
(c) (A) - (iv), (B) - (ii), (C) - (i), (D) - (iii)
(d) (A) - (ii), (B) - (i), (C) - (iv), (D) - (iii)

**40.** Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

**Assertion (A)** The stretching of a spring is determined by the shear modulus of the material of the spring.

**Reason (R)** A coil spring of copper has more tensile strength than a steel spring of same dimensions.

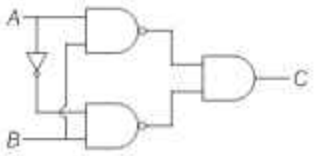
In the light of the above statements, choose the most appropriate answer from the options given below

(a) Both A and R are true and R is not the correct explanation of A.
(b) A is true but R is false.
(c) A is false but R is true.
(d) Both A and R are true and R is the correct explanation of A.

**41.** The area of a rectangular field (in $m^2$) of length 55.3m and breadth 25m after rounding off the value for correct significant digits is

(a) 1382 (b) 1382.5
(c) $14 \times 10^2$ (d) $138 \times 10^1$

**42.**




The truth table for the given logic circuit is

(a)

| A | B | C |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

(b)

| A | B | C |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |

A B C

(c)

| A | B | C |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

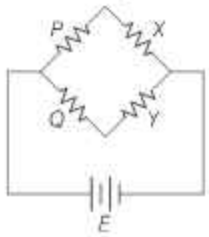
(d)

| A | B | C |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

**43.** The volume occupied by the molecules contained in 4.5 kg water at STP, if the intermolecular forces vanish away is

(a) $5.6 \times 10^3$ $m^3$ (b) $5.6 \times 10^{-3}$ $m^3$
(c) 5.6 $m^3$ (d) $5.6 \times 10^6$ $m^3$

**44.** A wheatstone bridge is used to determine the value of unknown resistance $X$ by adjusting the variable resistance $Y$ as shown in the figure. For the most precise measurement of $X$, the resistances $P$ and $Q$

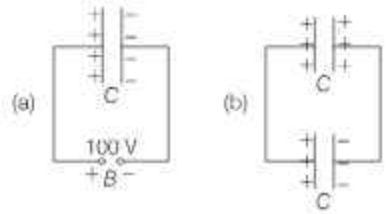



(a) should be approximately equal and are small
(b) should be very large and unequal
(c) do not play any significant role
(d) should be approximately equal to $2X$

**45.** A series $L$-$C$-$R$ circuit with inductance 10 H, capacitance 10 μF, resistance 50Ω is connected to an AC source of voltage, $V = 200 \sin(100\ t)$ volt. If the resonant frequency of the LCR circuit is $\nu_0$ and the frequency of the AC source is $\nu$, then

(a) $\nu_0 = \nu = \frac{50}{\pi}$ Hz (b) $\nu_0 = \frac{50}{\pi}$ Hz, $\nu = 50$ Hz
(c) $\nu = 100$ Hz; $\nu_0 = \frac{100}{\pi}$ Hz (d) $\nu_0 = \nu = 50$ Hz

**46.** A capacitor of capacitance $C = 900$ pF is charged fully by 100 V battery $B$ as shown in figure (a). Then, it is disconnected from the battery and connected to another uncharged capacitor of capacitance $C = 900$ pF as shown in figure (b). The electrostatic energy stored by the system (b) is




(a) $3.25 \times 10^{-6}$ J (b) $2.25 \times 10^{-6}$ J
(c) $1.5 \times 10^{-6}$ J (d) $4.5 \times 10^{-6}$ J

**47.** Two point charges $-q$ and $+q$ are placed at a distance of $L$, as shown in the figure.




The magnitude of electric field intensity at a distance $R(R >> L)$ varies as

(a) $\frac{1}{R^3}$ (b) $\frac{1}{R^4}$ (c) $\frac{1}{R^5}$ (d) $\frac{1}{R^2}$

**48.** A ball is projected with a velocity, 10 $ms^{-1}$, at an angle of 60° with the vertical direction. Its speed at the highest point of its trajectory will be

(a) $5\sqrt{3}$ $ms^{-1}$ (b) 5 $ms^{-1}$ (c) 10 $ms^{-1}$ (d) Zero

***49.** From Ampere's circuital law for a long straight wire of circular cross-section carrying a steady current, the variation of magnetic field in the inside and outside region of the wire is :

(a) a linearly increasing function of distance upto the boundary of the wire and then linearly decreasing for the outside region.

(b) a linearly increasing function of distance $r$ upto the boundary of the wire and then decreasing one with $1/r$ dependence for the outside region.

(c) a linearly decreasing function of distance upto the boundary of the wire and then a linearly increasing one for the outside region.

(d) uniform and remains constant for both the regions.

**50.** Two pendulums of length 121 cm and 100 cm start vibrating in phase. At some instant, the two are at their mean position in the same phase. The minimum number of vibrations of the shorter pendulum after which the two are again in phase at the mean position is

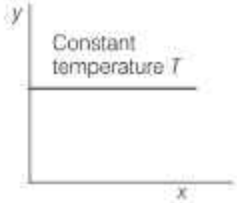
(a) 9 (b) 10
(c) 8 (d) 11

# CHEMISTRY

## Section-A

**51.** The given graph is a representation of kinetics of a reaction.

The $y$ and $x$ axes for zero and first order reactions, respectively are



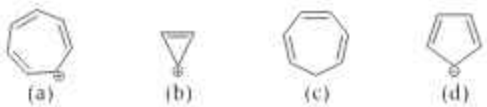

(a) zero order ($y$ = concentration and $x$ = time), first order ($y$ = rate constant and $x$ = concentration).

(b) zero order ($y$ = rate and $x$ = concentration), first order ($y = t_{1/2}$ and $x$ = concentration).

(c) zero order ($y$ = rate and $x$ = concentration), first order ($y$ = rate and $x = t_{1/2}$).

(d) zero order ($y$ = concentration and $x$ = time), first order ($y = t_{1/2}$ and $x$ = concentration).

**52.** Which compound amongst the following is not an aromatic compound?




**53.** $RMgX + CO_2 \xrightarrow[\text{ether}]{\text{Dry}} Y \xrightarrow{H_3O^+} RCOOH$

What is $Y$ in the above reaction ?

(a) $R_3CO^-Mg^+X$ (b) $RCOO^-X^+$
(c) $(RCOO)_2Mg$ (d) $RCOO^-Mg^+X$

**54.** Which of the following statement is not correct about diborane?

(a) The four terminal B—H bonds are 2-centre-2 electron bonds.

(b) The four terminal hydrogen atoms and the two boron atoms lie in one plane.

(c) Both the boron atoms are $sp^2$-hybridised.

(d) There are two 3-centre-2-electron bonds.

**55.** Gadolinium has a low value of third ionisation enthalpy because of

(a) high exchange enthalpy

(b) high electronegativity

(c) high basic character

(d) small size

***56.** Which one is not correct mathematical equation for Dalton's law of partial pressure? Here $p$ = total pressure of gaseous mixture

(a) $p = n_1\frac{RT}{V} + n_2\frac{RT}{V} + n_3\frac{RT}{V}$

(b) $p_i = x_i p$, where $p_i$ = partial pressure of $i$th gas $x_i$ = mole fraction of $i$th gas in gaseous mixture.

(c) $p_i = x_i p_i^0$, where $x_i$ = mole fraction of $i^{th}$ gas in gaseous mixture, $p_i^0$ = pressure of $i$th gas in pure state.

(d) $p = p_1 + p_2 + p_3$

**57.** Given below are two statements.

**Statement I** The boiling points of the following hydrides of group 16 elements increases in the order

$H_2O < H_2S < H_2Se < H_2Te$

**Statement II** The boiling points of these hydride increases with increase in molar mass.

In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both statement I and statement II are incorrect.
(b) Statement I is correct but statement II is incorrect.
(c) Statement I is incorrect but statement II correct.
(d) Both statement I and statement II are correct.

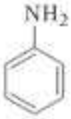
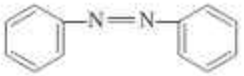
**58.** Given below are two statements.

**Statement I** The acidic strength of monosubstituted nitrophenol is higher than phenol because of electron withdrawing nitro group.

**Statement II** *o*-nitrophenol, *m*-nitrophenol and *p*-nitrophenol will have same acidic strength as they have one nitro group attached to the phenolic ring.

In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both statement I and statement II are incorrect.
(b) Statement I is correct but statement II is incorrect.
(c) Statement I is incorrect but statement II is correct.
(d) Both statement I and statement II are correct.

**59.** The Kjeldahl's method for the estimation of nitrogen can be used to estimate the amount of nitrogen in which one of the following compounds?

(a) 
(b) 
(c) 
(d) 

***60.** Match List-I with List-II.

| | List-I (Drug class) | | List-II (Drug molecule) |
|---|---|---|---|
| (A) | Antacids | (i) | Salvarsan |
| (B) | Antihistamines | (ii) | Morphine |
| (C) | Analgesics | (iii) | Cimetidine |
| (D) | Antimicrobials | (iv) | Seldane |

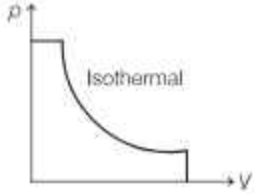
Choose the correct answer from the options given below
(a) (A) - (iii), (B) - (iv), (C) - (ii), (D) - (i)
(b) (A) - (i), (B) - (iv), (C) - (ii), (D) - (iii)
(c) (A) - (iv), (B) - (iii), (C) - (i), (D) - (ii)
(d) (A) - (iii), (B) - (ii), (C) - (iv), (D) - (i)

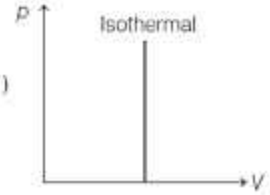
**61.** The IUPAC name of an element with atomic number 119 is
(a) unnilennium (b) ununnnium
(c) ununoctium (d) ununennium

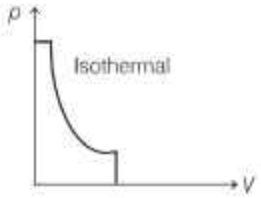
**62.** Amongst the following which one will have maximum 'lone pair - lone pair' electron repulsions?
(a) $IF_5$ (b) $SF_4$
(c) $XeF_2$ (d) $ClF_3$

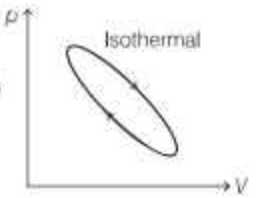
**63.** Which of the following sequence of reactions is suitable to synthesise chlorobenzene?
(a) Phenol, $NaNO_2$, HCl, CuCl
(b)  , HCl
(c)  , HCl, Heating
(d) Benzene, $Cl_2$, anhydrous $FeCl_3$

**64.** Which of the following *p*-*V* curve represents maximum work done?

(a) 

(b) 

(c) 

(d) 


**65.** Given below are half-cell reactions

$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$,

$E^\ominus_{Mn^{2+}/MnO_4^-} = -1.510$ V

$\frac{1}{2}O_2 + 2H^+ + 2e^- \longrightarrow H_2O$, $E^\ominus_{O_2/H_2O} = +1.223$ V

Will the permanganate ion, $MnO_4^-$ liberate $O_2$ from water in the presence of an acid?
(a) No, because $E^\ominus_{cell} = -0.287$ V
(b) Yes, because $E^\ominus_{cell} = +2.733$ V
(c) No, because $E^\ominus_{cell} = -2.733$ V
(d) Yes, because $E^\ominus_{cell} = +0.287$ V

**66.** The IUPAC name of the complex $[Ag(H_2O)_2][Ag(CN)_2]$ is
(a) diaquasilver (II) dicyanidoargentate (II)
(b) dicyanidosilver (I) diaquaargentate (I)
(c) diaquasilver (I) dicyanidoargentate (I)
(d) dicyanidosilver (II) diaquaargentate (II)

***67.** Identify the incorrect statement from the following.
(a) The oxidation number of K in $KO_2$ is + 4.
(b) Ionisation enthalpy of alkali metals decreases from top to bottom in the group.
(c) Lithium is the strongest reducing agent among the alkali metals.
(d) Alkali metals react with water to form their hydroxides.

**68.** What mass of 95% pure $CaCO_3$ will be required to neutralise 50 mL of 0.5 M HCl solution according to the following reaction?

$$CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$$

[Calculate upto second place of decimal point]

(a) 1.32 g (b) 3.65 g (c) 9.50 g (d) 1. 25 g

**69.** Given below are two statements.

**Statement I** Primary aliphatic amines react with $HNO_2$ to give unstable diazonium salts.

**Statement II** Primary aromatic amines react with $HNO_2$ to form diazonium salts which are stable even above 300 K.

In the light of the above statements, choose the most appropriate answer from the options given below.

(a) Both statement I and statement II are incorrect.
(b) Statement I is correct but statement II is incorrect.
(c) Statement I is incorrect but statement II is correct.
(d) Both statement I and statement II are correct.

***70.** Which statement regarding polymers is not correct?

(a) Fibers possess high tensile strength.
(b) Thermoplastic polymers are capable of repeatedly softening and hardening on heating and cooling respectively.
(c) Thermosetting polymers are reusable.
(d) Elastomers have polymer chains held together by weak intermolecular forces.

**71.** The incorrect statement regarding chirality is

(a) the product obtained by $S_N2$ reaction of haloalkane having chirality at the reactive site shows inversion of configuration.
(b) enantiomers are superimposable mirror images on each other.
(c) a racemic mixture shows zero optical rotation.
(d) $S_N1$ reaction yields 1 : 1 mixture of both enantiomers.

***72.** Given below are two statements.

**Statement I** In the coagulation of a negative sol, the flocculating power of the three given ions is in the order

$$Al^{3+} > Ba^{2+} > Na^+$$

**Statement II** In the coagulation of a positive sol, the flocculating power of the three given salts is in the order

$$NaCl > Na_2SO_4 > Na_3PO_4$$

In the light of the above statements, choose the most appropriate answer from the options given below.

(a) Both statement I and statement II are incorrect.
(b) Statement I is correct but statement II is incorrect.
(c) Statement I is incorrect but statement II is correct.
(d) Both statement I and statement II are correct.

**73.** Match List-I with List-II.

| | **List-I** (Hydrides) | | **List-II** (Nature) |
|---|---|---|---|
| (A) | $MgH_2$ | (i) | Electron precise |
| (B) | $GeH_4$ | (ii) | Electron deficient |
| (C) | $B_2H_6$ | (iii) | Electron rich |
| (D) | HF | (iv) | Ionic |

Choose the correct answer from the options given below

(a) (A) - (iii), (B) - (i), (C) - (ii), (D) - (iv)
(b) (A) - (i), (B) - (ii), (C) - (iv), (D) - (iii)
(c) (A) - (ii), (B) - (iii), (C) - (iv), (D) - (i)
(d) (A) - (iv), (B) - (i), (C) - (ii), (D) - (iii)

***74.** Match List-I with List-II.

| | **List-I** | | **List-II** |
|---|---|---|---|
| (A) | Li | (i) | Absorbent for carbon dioxide |
| (B) | Na | (ii) | Electrochemical cells |
| (C) | KOH | (iii) | Coolant in fast breeder reactors |
| (D) | Cs | (iv) | Photoelectric cell |

Choose the correct answer from the options given below

(a) (A) - (iii), (B) - (iv), (c) - (ii), (D) - (i)
(b) (A) - (i), (B) - (iii), (c) - (iv), (D) - (ii)
(c) (A) - (ii), (B) - (iii), (c) - (i), (D) - (iv)
(d) (A) - (iv), (B) - (i), (c) - (iii), (D) - (ii)

***75.** Given below are two statements one is labelled as Assertion (A) and the other is labelled as Reason (R).

**Assertion (A)** In a particular point defect, an ionic solid is electrically neutral, even if few of its cations are missing from its unit cells.

**Reason (R)** In an ionic solid, Frenkel defect arises due to dislocation of cation from its lattice site to interstitial site, maintaining overall electrical neutrality.

In the light of the above statements, choose the most appropriate answer from the options given below.

(a) Both A and R are correct but R is not the correct explanation of A.
(b) A is correct but R is not correct.
(c) A is not correct but R is correct.
(d) Both A and R are correct and R is the correct explanation of A.

**76.** The incorrect statement regarding enzymes is

(a) like chemical catalysts enzymes reduce the activation energy of bio processes.
(b) enzymes are polysaccharides.
(c) enzymes are very specific for a particular reaction and substrate.
(d) enzymes are biocatalysts.

**77.** Match List-I with List-II.

| | **List-I** (Products formed) | | **List-II** (Reaction of carbonyl compound with) |
|---|---|---|---|
| (A) | Cyanohydrin | (i) | $NH_2OH$ |
| (B) | Acetal | (ii) | $RNH_2$ |
| (C) | Schiff's base | (iii) | Alcohol |
| (D) | Oxime | (iv) | HCN |

Choose the correct answer from the options given below

(a) (A) - (ii), (B) - (iii), (C) - (iv), (D) - (i)

(b) (A) - (i), (B) - (iii), (C) - (ii), (D) - (iv)

(c) (A) - (iv), (B) - (iii), (C) - (ii), (D) - (i)

(d) (A) - (iii), (B) - (iv), (C) - (ii), (D) - (i)

**78.** In one molal solution that contains 0.5 mole of a solute, there is

(a) 500 g of solvent

(b) 100 mL of solvent

(c) 1000 g of solvent

(d) 500 mL of solvent

**79.** Choose the correct statement.

(a) Diamond is covalent and graphite is ionic.

(b) Diamond is $sp^3$-hybridised and graphite is $sp^2$-hybridised.

(c) Both diamond and graphite are used as dry lubricants.

(d) Diamond and graphite have two dimensional network.

**80.** Which amongst the following is incorrect statement?

(a) $C_2$ molecule has four electrons in its two degenerate π-molecular orbitals.

(b) $H_2^+$ ion has one electron.

(c) $O_2^+$ ion is diamagnetic.

(d) The bond orders of $O_2^+$, $O_2$, $O_2^-$ and $O_2^{2-}$ are 2.5, 2, 1.5 and 1 respectively.

**81.** Given below are two statements one is labelled as Assertion (A) and the other is labelled as Reason (R).

**Assertion (A)** ICl is more reactive than $I_2$.

**Reason (R)** I—Cl bond is weaker than I—I bond.

In the light of the above statements, choose the most appropriate answer from the options given below .

(a) Both A and R are correct but R is not the correct explanation of A.

(b) A is correct but R is not correct.

(c) A is not correct but R is correct.

(d) Both A and R are correct and R is the correct explanation of A.

**82.** Given below are two statements.

**Statement I** The boiling points of aldehydes and ketones are higher than hydrocarbons of comparable molecular masses because of weak molecular association in aldehydes and ketones due to dipole-dipole interactions.

**Statement II** The boiling points of aldehydes and ketones are lower than the alcohols of similar molecular masses due to the absence of H-bonding.

In the light of the above statements, choose the most appropriate answer from the options given below

(a) Both statement I and statement II are incorrect.

(b) Statement I is correct but statement II is incorrect.

(c) Statement I is incorrect but statement II is correct.

(d) Both statement I and statement II are correct.

**83.** The pH of the solution containing 50 mL each of 0.10 M sodium acetate and 0.01 M acetic acid is [Given $pK_a$ of $CH_3COOH = 4.57$]

(a) 3.57 (b) 4.57 (c) 2.57 (d) 5.57

**84.** Identify the incorrect statement from the following.

(a) All the five 4*d*-orbitals have shapes similar to the respective 3*d*-orbitals.

(b) In an atom, all the five 3*d*-orbitals are equal in energy in free state.

(c) The shapes of $d_{xy}$, $d_{yz}$, and $d_{zx}$ orbitals are similar to each other; and $d_{x^2-y^2}$ and $d_{z^2}$ are similar to each other.

(d) All the five 5*d*-orbitals are different in size when compared to the respective 4*d*-orbitals.

**85.** At 298 K, the standard electrode potentials of $Cu^{2+}/Cu$, $Zn^{2+}/Zn$, $Fe^{2+}/Fe$ and $Ag^+/Ag$ are 0.34 V, −0.76 V, −0.44 V and 0.80 V, respectively. On the basis of standard electrode potential, predict which of the following reaction can not occur?

(a) $CuSO_4(aq) + Fe(s) \longrightarrow FeSO_4(aq) + Cu(s)$

(b) $FeSO_4(aq) + Zn(s) \longrightarrow ZnSO_4(aq) + Fe(s)$

(c) $2CuSO_4(aq) + 2Ag(s) \longrightarrow 2Cu(s) + Ag_2SO_4(aq)$

(d) $CuSO_4(aq) + Zn(s) \longrightarrow ZnSO_4(aq) + Cu(s)$

## Section-B

**86.** The order of energy absorbed which is responsible for the colour of complexes

(A) $[Ni(H_2O)_2(en)_2]^{2+}$

(B) $[Ni(H_2O)_4(en)]^{2+}$ and

(C) $[Ni(en)_3]^{2+}$ is

(a) (C) > (B) > (A) (b) (C) > (A) > (B)

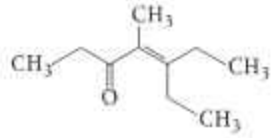
(c) (B) > (A) > (C) (d) (A) > (B) > (C)

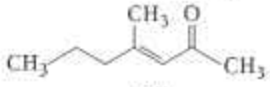
***87.** A 10.0 L flask contains 64 g of oxygen at 27°C. (Assume $O_2$ gas is behaving ideally). The pressure inside the flask in bar is

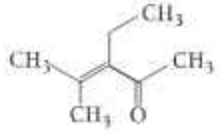
(Given, $R = 0.0831$ L bar $K^{-1}$ $mol^{-1}$)

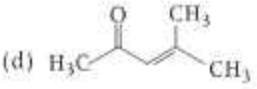
(a) 498.6 (b) 49.8 (c) 4.9 (d) 2.5

**88.** Which one of the following is not formed when acetone reacts with 2-pentanone in the presence of dilute NaOH followed by heating?

(a) 

(b) 

(c) 

(d) 

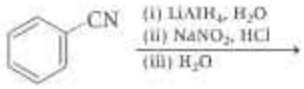
**89.** Find the emf of the cell in which the following reaction takes place at 298 K

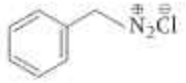
$Ni(s) + 2Ag^+(0.001\ M) \longrightarrow Ni^{2+}(0.001\ M) + 2Ag(s)$

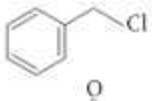
(Given that, $E^\circ_{cell} = 10.5$ V, $\frac{2.303\,RT}{F} = 0.059$ at 298 K)

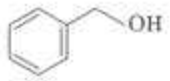
(a) 1.385 V (b) 0.9615 V
(c) 1.05 V (d) 1.0385 V

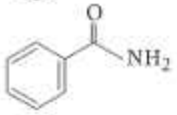
**90.** The product formed from the following reaction sequence is




(a) 

(b) 

(c) 

(d) 


***91.** Copper crystallises in fcc unit cell with cell edge length of $3.608 \times 10^{-8}$ cm. The density of copper is 8.92 g $cm^{-3}$. Calculate the atomic mass of copper.

(a) 31.55 u (b) 60 u
(c) 65 u (d) 63.1 u

**92.** $3O_2(g) \rightleftharpoons 2O_3(g)$

For the above reaction at 298 K, $K_C$ is found to be $3.0 \times 10^{-59}$. If the concentration of $O_2$ at equilibrium is 0.040 M then concentration of $O_3$ in M is

(a) $1.9 \times 10^{-63}$ (b) $2.4 \times 10^{31}$
(c) $1.2 \times 10^{21}$ (d) $4.38 \times 10^{-32}$

**93.** Given below are two statements.

**Statement I** In Lucas test, primary, secondary and tertiary alcohols are distinguished on the basis of their reactivity with conc. HCl + $ZnCl_2$, known as Lucas reagent.

**Statement II** Primary alcohols are most reactive and immediately produce turbidity at room temperature on reaction with Lucas reagent.

In the light of the above statements, choose the most appropriate answer from the options given below.

(a) Both statement I and statement II are incorrect.
(b) Statement I is correct but statement II is incorrect.
(c) Statement I is incorrect but statement II is correct.
(d) Both statement I and statement II are correct.

**94.** In the neutral or faintly alkaline medium, $KMnO_4$ oxidises iodide into iodate. The change in oxidation state of manganese in this reaction is from

(a) + 6 to + 4 (b) + 7 to + 3
(c) + 6 to + 5 (d) + 7 to + 4

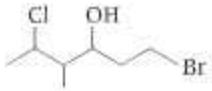
***95.** Match List-I with List-II.

| | List-I (Ores) | | List-II (Composition) |
|---|---|---|---|
| (A) | Haematite | (i) | $Fe_3O_4$ |
| (B) | Magnetite | (ii) | $ZnCO_3$ |
| (C) | Calamine | (iii) | $Fe_2O_3$ |
| (D) | Kaolinite | (iv) | $[Al_2(OH)_4Si_2O_5]$ |

Choose the correct answer from the options given below

(a) (A) - (iii), (B) - (i), (C) - (ii), (D) - (iv)
(b) (A) - (iii), (B) - (i), (C) - (iv), (D) - (ii)
(c) (A) - (i), (B) - (iii), (C) - (ii), (D) - (iv)
(d) (A) - (i), (B) - (ii), (C) - (iii), (D) - (iv)

**96.** The correct IUPAC name of the following compound is




(a) 6-bromo-2-chloro-4-methylhexan-4-ol
(b) 1-bromo-4-methyl-5-chlorohexan-3-ol
(c) 6-bromo-4-methyl-2-chlorohexan-4-ol
(d) 1-bromo-5-chloro-4-methyhexan-3-ol

**97.** If radius of second Bohr orbit of the $He^+$ ion is 105.8 pm, what is the radius of third Bohr orbit of $Li^{2+}$ ion?

(a) 15.87 pm (b) 1.587 pm
(c) 158.7 Å (d) 158.7 pm

**98.** Compound $X$ on reaction with $O_3$ followed by $Zn/H_2O$ gives formaldehyde and 2-methyl propanal as products. The compound $X$ is

(a) 2-methylbut-1-ene (b) 2-methylbut-2-ene
(c) Pent-2-ene (d) 3-methylbut-1-ene

**99.** For a first order reaction $A \rightarrow$ products, initial concentration of $A$ is 0.1 M, which becomes 0.001 M after 5 minutes. Rate constant for the reaction in $min^{-1}$ is

(a) 0.9212 (b) 0.4606
(c) 0.2303 (d) 1.3818

***100.** The pollution due to oxides of sulphur gets enhanced due to the presence of

(I) particulate matter
(II) ozone
(III) hydrocarbons
(IV) hydrogen peroxide

Choose the most appropriate answer from the options given below

(a) (I),(II),(IV) only
(b) (II), (III), (IV) only
(c) (I), (III), (IV) only
(d) (I), (IV) only

# BIOLOGY

## Section-A

**101.** Which of the following is not a method of *ex situ* conservation?
(a) National parks (b) Micropropagation
(c) Cryopreservation (d) *In vitro* fertilisation

**102.** Given below are two statements

**Statement I** The primary $CO_2$ acceptor in $C_4$ plants is phosphoenolpyruvate and is found in the mesophyll cells.

**Statement II** Mesophyll cells of $C_4$ plants lack RuBisCO enzyme.

In the light of the above statements, choose the correct answer from the options given below.
(a) Both statement I and statement II are incorrect.
(b) Statement I is correct, but statement II is incorrect.
(c) Statement I is incorrect, but statement II is correct.
(d) Both statement I and statement II are correct.

**103.** XO type of sex-determination can be found in
(a) birds (b) grasshoppers
(c) monkeys (d) *Drosophila*

**104.** In old trees, the greater part of secondary xylem is dark brown and resistant to insect attack due to
I. secretion of secondary metabolites and their deposition in the lumen of vessels.
II. deposition of organic compounds like tannins and resins in the central layers of stem.
III. deposition of suberin and aromatic substances in the outer layer of stem.
IV. deposition of tannins, gum, resin and aromatic substances in the peripheral layers of stem.
V. presence of parenchyma cells, functionally active xylem elements and essential oils.

Choose the correct answer from the options given below.
(a) (III) and (IV) (b) (IV) and (V)
(c) (II) and (IV) (d) (I) and (II)

***105.** Which of the following is not observed during apoplastic pathway?
(a) The movement does not involve crossing of cell membrane
(b) The movement is aided by cytoplasmic streaming
(c) Apoplast is continuous and does not provide any barrier to water movement
(d) Movement of water occur through intercellular spaces and wall of the cells

**106.** Given below are two statements, one is labelled as Assertion (A) and the other is labelled as Reason (R)

**Assertion (A)** Polymerase chain reaction is used in DNA amplification.

**Reason (R)** The ampicillin resistant gene is used as a selectable marker to check transformation.

In the light of the above statements, choose the correct answer from the options given below.
(a) Both (A) and (R) are correct, but (R) is not the correct explanation of (A).
(b) (A) is correct, but (R) is not correct.
(c) (A) is not correct, but (R) is correct.
(d) Both (A) and (R) are correct and (R) is the correct explanation of (A).

**107.** Which one of the following statement is not true regarding gel electrophoresis technique?
(a) The separated DNA fragments are stained by using ethidium bromide
(b) The presence of chromogenic substrate gives blue coloured DNA bands on the gel.
(c) Bright orange coloured bands of DNA can be observed in the gel when exposed to UV light
(d) The process of extraction of separated DNA strands from gel is called elution

**108.** The gaseous plant growth regulator is used in plants to
(a) promote root growth and root hair formation to increase the absorption surface
(b) help overcome apical dominance
(c) kill dicotyledonous weeds in the fields
(d) speed up the malting process

**109.** What amount of energy is released from glucose during lactic acid fermentation?
(a) More than 18% (b) About 10%
(c) Less than 7% (d) Approximately 15%

**110.** Read the following statements about the vascular bundles.
I. In roots, xylem and phloem in a vascular bundle are arranged in an alternate manner along the different radii.
II. Conjoint closed vascular bundles do not possess cambium.
III. In open vascular bundles, cambium is present in between xylem and phloem.
IV. The vascular bundles of dicotyledonous stem possess endarch protoxylem.
V. In monocotyledonous root, usually there are more than six xylem bundles present

Choose the correct answer from the options given below
(a) II, III, IV and V (b) I, II, III and IV
(c) I, III, IV and V (d) I, II, III, IV and V

**111.** Identify the correct set of statements.
I. The leaflets are modified into pointed hard thorns in *Citrus* and *Bougainvillea*.
II. Axillary buds form slender and spirally coiled tendrils in cucumber and pumpkin.
III. Stem is flattened and fleshy in *Opuntia* and modified to perform the function of leaves.

IV. *Rhizophora* shows vertically upward growing roots that help to get oxygen for respiration.
V. Subaerially growing stems in grasses and strawberry help in vegetative propagation.

Choose the correct answer from the options given below.
(a) I and IV (b) II, III, IV and V
(c) I, II, IV and V (d) II and III

**112.** Which one of the following plants does not show plasticity?
(a) Coriander (b) Buttercup (c) Maize (d) Cotton

**113.** Given below are two statements,

**Statement I** Cleistogamous flowers are invariably autogamous.

**Statement II** Cleistogamy is disadvantageous as there is no chance for cross-pollination.

In the light of the above statements, choose the correct answer from the options given below.
(a) Both statement I and statement II are incorrect
(b) Statement I is correct, but statement II is incorrect
(c) Statement I is incorrect, but statement II is correct
(d) Both statement I and statement II are correct

***114.** 'Girdling Experiment' was performed by plant physiologists to identify the plant tissue through which
(a) food is transported
(b) for both water and food transportation
(c) osmosis is observed
(d) water is transported

**115.** Hydrocolloid carrageen is obtained from
(a) Phaeophyceae and Rhodophyceae
(b) Rhodophyceae only
(c) Phaeophyceae only
(d) Chlorophyceae and Phaeophyceae

**116.** Given below are two statements,

**Statement I** Mendel studied seven pairs of contrasting traits in pea plants and proposed the law of inheritance.

**Statement II** Seven characters examined by Mendel in his experiment on pea plants were seed shape and colour, flower colour, pod shape and colour, flower position and stem height.

In the light of the above statements, choose the correct answer from the options given below
(a) Both statement I and statement II are incorrect
(b) Statement I is correct, but statement II is incorrect
(c) Statement I is incorrect but statement II is correct
(d) Both statement I and statement II are correct

**117.** Production of cucumber has increased manifold in recent years. Application of which of the following phytohormones has resulted in this increased yield as the hormone is known to produce female flowers in the plants ?
(a) Gibberellin (b) Ethylene
(c) Cytokinin (d) ABA

**118.** The process of translation of *m*RNA to proteins begins as soon as
(a) the larger subunit of ribosome encounters *m*RNA
(b) both the subunits join together to bind with *m*RNA
(c) the *t*RNA is activated and the larger subunit of ribosome encounters *m*RNA
(d) the small subunit of ribosome encounters *m*RNA

**119.** Habitat loss and fragmentation, overexploitation, alien-species invasion and co-extinction are causes for
(a) competition (b) biodiversity loss
(c) natality (d) population explosion

**120.** Read the following statements and choose the set of correct statements.
I. Euchromatin is loosely packed chromatin.
II. Heterochromatin is transcriptionally active.
III. Histone octamer is wrapped by negatively charged DNA in nucleosome.
IV. Histones are rich in lysine and arginine.
V. A typical nucleosome contains 400 bp of DNA helix.

Choose the correct answer from the options given below.
(a) I, III and IV (b) II and V
(c) I, III and V (d) II, IV and V

**121.** Which of the following is incorrectly matched?
(a) *Ulothrix*—Mannitol
(b) *Porphyra*—Floridean starch
(c) *Volvox*—Starch
(d) *Ectocarpus*—Fucoxanthin

**122.** The device which can remove particulate matter present in the exhaust from a thermal power plant is
(a) incinerator (b) electrostatic precipitator
(c) catalytic convertor (d) STP

**123.** DNA polymorphism forms the basis of
(a) DNA fingerprinting
(b) both genetic mapping and DNA fingerprinting
(c) translation
(d) genetic mapping

**124.** Which one of the following statements cannot be connected to predation?
(a) It might lead to extinction of a species
(b) Both the interacting species are negatively impacted
(c) It is necessitated by nature to maintain the ecological balance
(d) It helps in maintaining species diversity in a community

**125.** Which one of the following never occurs during mitotic cell division?
(a) Movement of centrioles towards opposite poles
(b) Pairing of homologous chromosomes
(c) Coiling and condensation of the chromatids
(d) Spindle fibres attach to kinetochores of chromosomes

**126.** Exoskeleton of arthropods is composed of
(a) cellulose (b) chitin
(c) glucosamine (d) cutin

**127.** What is the net gain of ATP when each molecule of glucose is converted to two molecules of pyruvic acid?
(a) 6 (b) 2
(c) 8 (d) 4

**128.** Which one of the following plants shows vexillary aestivation and diadelphous stamens?
(a) *Pisum sativum* (b) *Allium cepa*
(c) *Solanum nigrum* (d) *Colchicum autumnale*

**129.** Identify the incorrect statement related to pollination.
(a) Pollination by wind is more common amongst abiotic pollination
(b) Flowers produce foul odours to attract flies and beetles to get pollinated
(c) Moths and butterflies are the most dominant pollinating agents among insects
(d) Pollination by water is quite rare in flowering plants

**130.** Given below are two statements.
**Statement I** Decomposition is a process in which the detritus is degraded into simpler substances by microbes.
**Statement II** Decomposition is faster if the detritus is rich in lignin and chitin.
In the light of the above statements, choose the correct answer from the options given below.
(a) Both statement I and statement II are incorrect
(b) Statement I is correct, but statement II is incorrect
(c) Statement I is incorrect, but statement II is correct
(d) Both statement I and statement II are correct

**131.** The appearance of recombination nodules on homologous chromosomes during meiosis characterises
(a) bivalent
(b) sites at which crossing over occurs
(c) terminalisation
(d) synaptonemal complex

***132.** Which one of the following produces nitrogen-fixing nodules on the roots of *Alnus* ?
(a) *Frankia* (b) *Rhodospirillum*
(c) *Beijerinckia* (d) *Rhizobium*

**133.** Which one of the following is not true regarding the release of energy during ATP synthesis through chemiosmosis? It involves
(a) breakdown of electron gradient
(b) movement of protons across the membrane to the stroma
(c) reduction of $NADP^+$ to $NADPH_2$ on the stroma side of the membrane
(d) breakdown of proton gradient

**134.** The flowers are zygomorphic in
I. Mustard II. Gulmohar
III. *Cassia* IV. *Datura*
V. Chilly
Choose the correct answer from the options given below.
(a) (II) and (III) (b) (IV) and (V)
(c) (III), (IV) and (V) (d) (I), (II) and (III)

***135.** Match List-I with List-II.

| | List-I | | List-II |
|---|---|---|---|
| A. | Manganese | (i) | Activates the enzyme catalase |
| B. | Magnesium | (ii) | Required for pollen germination |
| C. | Boron | (iii) | Activates enzymes of respiration |
| D. | Iron | (iv) | Functions in splitting of water during photosynthesis |

Choose the correct answer from the options given below.
(a) (A) - (iv), (B) - (iii), (C) - (ii), (D) - (i)
(b) (A) - (iv), (B) - (i), (C) - (ii), (D) - (iii)
(c) (A) - (iii), (B) - (i), (C) - (ii), (D) - (iv)
(d) (A) - (iii), (B) - (iv), (C) - (i), (D) - (ii)

## Section-B

***136.** Addition of more solutes in a given solution will
(a) lower its water potential
(b) make its water potential zero
(c) not affect the water potential at all
(d) raise its water potential

**137.** Read the following statements on lipids and find out correct set of statements
I. Lecithin found in the plasma membrane is a glycolipid.
II. Saturated fatty acids possess one or more C = C bonds.
III. Gingelly oil has lower melting point, hence remains as oil in winter.
IV. Lipids are generally insoluble in water, but soluble in some organic solvents.
V. When fatty acid is esterified with glycerol, monoglycerides are formed.
Choose the correct answer from the options given below.
(a) I, IV and V (b) III, IV and V
(c) I, II and IV (d) I, II and III

**138.** What is the role of large bundle sheath cells found around the vascular bundles in $C_4$ plants?
(a) To increase the number of chloroplast for the operation of Calvin cycle
(b) To enable the plant to tolerate high temperature
(c) To protect the vascular tissue from high light intensity
(d) To provide the site for photorespiratory pathway

*139. The entire fleet of buses in Delhi were converted to CNG from diesel. In reference to this, which one of the following statements is false?
(a) The same diesel engine is used in CNG buses making the cost of conversion low
(b) It is cheaper than diesel
(c) It cannot be adulterated like diesel
(d) CNG burns more efficiently than diesel

140. Transposons can be used during which one of the following ?
(a) Gene silencing (b) Autoradiography
(c) Gene sequencing (d) Polymerase chain reaction

141. Match List-I with List-II.

| | List-I | | List-II |
|---|---|---|---|
| A. | Metacentric chromosome | (i) | Centromere situated close to the end forming one extremely short and one very long arms |
| B. | Acrocentric chromosome | (ii) | Centromere at the terminal end |
| C. | Sub-metacentric | (iii) | Centromere in the middle forming two equal arms of chromosomes |
| D. | Telocentric chromosome | (iv) | Centromere slightly away from the middle forming one shorter arm and one longer arm |

Choose the correct answer from the options given below.
(a) (A) - (i), (B) - (iii), (C) - (ii), (D) - (iv)
(b) (A) - (ii), (B) - (iii), (C) - (iv), (D) - (i)
(c) (A) - (i), (B) - (ii), (C) - (iii), (D) - (iv)
(d) (A) - (iii), (B) - (i), (C) - (iv), (D) - (ii)

142. Match the plant with the kind of life cycle it exhibits.

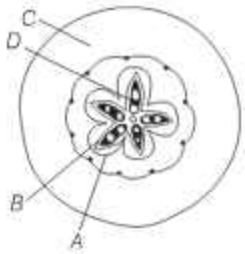
| | List-I | | List-II |
|---|---|---|---|
| A. | *Spirogyra* | (i) | Dominant diploid sporophyte vascular plant, with highly reduced male or female gametophyte |
| B. | Fern | (ii) | Dominant haploid free-living gametophyte |
| C. | *Funaria* | (iii) | Dominant diploid sporophyte alternating with reduced gametophyte called prothallus |
| D. | *Cycas* | (iv) | Dominant haploid leafy gametophyte alternating with partially dependent multicellular sporophyte |

Choose the correct answer from the options given below.
(a) (A) - (ii), (B) - (iii), (C) - (iv), (D) - (i)
(b) (A) - (iii), (B) - (iv), (C) - (i), (D) - (ii)
(c) (A) - (ii), (B) - (iv), (C) - (i), (D) - (iii)
(d) (A) - (iv), (B) - (i), (C) - (ii), (D) - (iii)

143. Which of the following occurs due to the presence of autosome linked dominant trait?
(a) Myotonic dystrophy (b) Haemophilia
(c) Thalessemia (d) Sickle cell anaemia

144. The anatomy of spring wood shows some peculiar features. Identify the correct set of statement about springwood.
I. It is also called as the early wood.
II. In spring season cambium produces xylem elements with narrow vessels.
III. It is light in colour.
IV. The springwood along with autumnwood shows alternate concentric rings forming annual rings.
V. It has lower density.
Choose the correct answer from the options given below.
(a) I, III, IV and V (b) I, II and IV
(c) III, IV and V (d) I, II, IV and V

145. Which part of the fruit, labelled in the given figure makes it a false fruit?




(a) B → Endocarp (b) C → Thalamus
(c) D → Seed (d) A → Mesocarp

146. In the following palindromic base sequences of DNA, which one can be cut easily by particular restriction enzyme?
(a) 5′ G A A T T C 3′; 3′ C T T A A G 5′
(b) 5′ C T C A G T 3′; 3′ G A G T C A 5′
(c) 5′ G T A T T C 3′; 3′ C A T A A G 5′
(d) 5′ G A T A C T 3′; 3′ C T A T G A 5′

147. While explaining interspecific interaction of population, (+) sign is assigned for beneficial interaction, (−) sign is assigned for detrimental interaction and (0) for neutral interaction. Which of the following interactions can be assigned (+) for one species and (−) for another species involved in the interaction ?
(a) Amensalism (b) Commensalism
(c) Competition (d) Predation

*148. Which one of the following will accelerate phosphrus cycle?
(a) Volcanic activity (b) Weathering of rocks
(c) Rainfall and storms (d) Burning of fossil fuels

149. Given below are two statements, one is labelled as Assertion (A) and the other is labelled as Reason (R).
**Assertion (A)** Mendel's law of independent assortment does not hold good for the genes that are located closely on the same chromosome.
**Reason (R)** Closely located genes assort independently.

In the light of the above statements, choose the correct answer from the options given below.
(a) Both (A) and (R) are correct, but (R) is not the correct explanation of (A)
(b) (A) is correct, but (R) is not correct
(c) (A) is not correct, but (R) is correct
(d) Both (A) and (R) are correct and (R) is the correct explanation of (A)

**150.** If a geneticist uses the blind approach for sequencing the whole genome of an organism, followed by assignment of function to different segments, the methodology adopted by him is called as,
(a) gene mapping
(b) expressed sequence tags
(c) bioinformatics
(d) sequence annotation

# ZOOLOGY

## Section-A

**151.** Given below are two statements.

**Statement I** Autoimmune disorder is a condition where body defence mechanism recognises its own cells as foreign bodies.

**Statement II** Rheumatoid arthritis is a condition where body does not attack self cells.

In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both statement I and statement II are incorrect
(b) Statement I is correct, but statement II is incorrect
(c) Statement I is incorrect, but statement II is correct
(d) Both statement I and statement II are correct

**152.** Given below are two statements.

**Statement I** The coagulum is formed of network of threads called thrombins.

**Statement II** Spleen is the graveyard of erythrocytes.

In the light of the above statements, choose the most appropriate answer from the options given below
(a) Both statement I and statement II are incorrect
(b) Statement I is correct, but statement II is incorrect
(c) Statement I is incorrect, but statement II is correct
(d) Both statement I and statement II are correct

**153.** Identify the asexual reproductive structure associated with *Penicillium*.
(a) Conidia (b) Gemmules (c) Buds (d) Zoospores

**154.** Given below are two statements.

**Statement I** The release of sperms into the seminiferous tubules is called spermiation.

**Statement II** Spermiogenesis is the process of formation of sperms from spermatogonia.

In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both statement I and statement II are incorrect
(b) Statement I is correct, but statement II is incorrect
(c) Statement I is incorrect, but statement II is correct
(d) Both statement I and statement II are correct

**155.** Under normal physiological condition in human being every 100 mL of oxygenated blood can deliver ..... mL of $O_2$ to the tissues.
(a) 5 mL (b) 4 mL (c) 10 mL (d) 2 mL

**156.** *In-situ* conservation refers to
(a) conserve only high risk species
(b) conserve only endangered species
(c) conserve only extinct species
(d) protect and conserve the whole ecosystem

**157.** Natural selection where more individuals acquire specific character value other than the mean character value, leads to
(a) directional change
(b) disruptive change
(c) random change
(d) stabilising change

***158.** Breeding crops with higher levels of vitamins and minerals or higher proteins and healthier fats is called
(a) bioremediation (b) biofortification
(c) bioaccumulation (d) biomagnification

**159.** Which of the following is present between the adjacent bones of the vertebral column?
(a) Cartilage (b) Areolar tissue
(c) Smooth muscle (d) Intercalated discs

**160.** Nitrogenous waste is excreted in the form of pellet or paste by
(a) *Salamandra* (b) *Hippocampus*
(c) *Pavo* (d) *Ornithorhynchus*

**161.** Which of the following statements with respect to endoplasmic reticulum is incorrect?
(a) SER is devoid of ribosomes
(b) In prokaryotes, only RER are present
(c) SER are the sites for lipid synthesis
(d) RER has ribosomes attached to ER

**162.** Which of the following statements are true for spermatogenesis but do not hold true for oogenesis?
I. It results in the formation of haploid gametes.
II. Differentiation of gamete occurs after the completion of meiosis.
III. Meiosis occurs continuously in a mitotically dividing stem cell population.
IV. It is controlled by the Luteinising hormone (LH) and Follicle Stimulating Hormone (FSH) secreted by the anterior pituitary.
V. It is initiated at puberty.

Choose the most appropriate answer from the options given below.
(a) II,and III (b) II, IV and V
(c) II, III and V (d) III and V

**163.** Given below are two statements, one is labelled as Assertion (A) and the other is labelled as Reason (R).
**Assertion (A)** All vertebrates are chordates but all chordates are not vertebrates.
**Reason (R)** Notochord is replaced by vertebral column in the adult vertebrates.
In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both (A) and (R) are correct, but (R) is not the correct explanation of (A)
(b) (A) is correct, but (R) is not correct
(c) (A) is not correct, but (R) is correct
(d) Both (A) and (R) are correct and (R) is the correct explanation of (A)

**164.** In which of the following animals, digestive tract has additional chambers like crop and gizzard?
(a) *Bufo, Balaenoptera, Bungarus*
(b) *Catla, Columba, Crocodylus*
(c) *Pavo, Psittacula, Corvus*
(d) *Corvus, Columba, Chameleon*

**165.** Given below are two statements, one is labelled as Assertion (A) and the other is labelled as Reason (R).
**Assertion (A)** Osteoporosis is characterised by decreased bone mass and increased chances of fractures.
**Reason (R)** Common cause of osteoporosis is increased levels of oestrogen.
In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both (A) and (R) are correct, but (R) is not the correct explanation of (A)
(b) (A) is correct, but (R) is not correct
(c) (A) is not correct, but (R) is correct
(d) Both (A) and (R) are correct and (R) is the correct explanation of (A)

**166.** In gene therapy of Adenosine Deaminase (ADA) deficiency, the patient requires periodic infusion of genetically engineered lymphocytes because
(a) Gene isolated from marrow cells producing ADA is introduced into cells at embryonic stages
(b) Lymphocytes from patient's blood are grown in culture, outside the body
(c) Genetically engineered lymphocytes are not immortal cells
(d) Retroviral vector is introduced into these lymphocytes

**167.** Select the incorrect statement with reference to mitosis.
(a) Spindle fibres attach to centromere of chromosomes
(b) Chromosomes decondense at telophase
(c) Splitting of centromere occurs at anaphase
(d) All the chromosomes lie at the equator at metaphase

**168.** In the taxonomic categories, which hierarchical arrangement in ascending order is correct in case of animals ?
(a) Kingdom, Class, Phylum, Family, Order, Genus, Species
(b) Kingdom, Order, Class, Phylum, Family, Genus, Species
(c) Kingdom, Order, Phylum, Class, Family, Genus, Species
(d) Kingdom, Phylum, Class, Order, Family, Genus, Species

**169.** Given below are two statements.
**Statement I** Restriction endonucleases recognise specific sequence to cut DNA known as palindromic nucleotide sequence.
**Statement II** Restriction endonucleases cut the DNA strand a little away from the centre of the palindromic site.
In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both statement I and statement II are incorrect.
(b) Statement I is correct, but statement II is incorrect.
(c) Statement I is incorrect, but statement II is correct.
(d) Both statement I and statement II are correct.

**170.** Given below are two statements.
**Statement I** *Mycoplasma* can pass through less than 1 micron filter size.
**Statement II** *Mycoplasma* are bacteria with cell wall.
In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both statement I and statement II are incorrect
(b) Statement I is correct, but statement II is incorrect
(c) Statement I is incorrect, but statement II is correct
(d) Both statement I and statement II are correct

**171.** Which of the following is not a connective tissue?
(a) Adipose tissue (b) Cartilage
(c) Neuroglia (d) Blood

**172.** Lippes loop is a type of contraceptive used as
(a) vault barrier (b) non-medicated IUD
(c) copper releasing IUD (d) cervical barrier

**173.** At which stage of life, the oogenesis process is initiated?
(a) Embryonic development stage
(b) Birth
(c) Adult
(d) Puberty

**174.** Identify the microorganism which is responsible for the production of an immunosuppressive molecule cyclosporin-A.
(a) *Clostridium butylicum* (b) *Aspergillus niger*
(c) *Streptococcus cerevisiae* (d) *Trichoderma polysporum*

***175.** Given below are two statements.
**Statement I** Fatty acids and glycerols cannot be absorbed into the blood.
**Statement II** Specialised lymphatic capillaries called lacteals carry chylomicrons into lymphatic vessels and ultimately into the blood.

In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both statement I and statement II are incorrect
(b) Statement I is correct, but statement II is incorrect
(c) Statement I is incorrect, but statement II is correct
(d) Both statement I and statement II are correct

**176.** In an *E.coli* strain, *i* gene gets mutated and its product cannot bind the inducer molecule. If growth medium is provided with lactose, what will be the outcome?
(a) *z*, *y* , a genes will be transcribed
(b) *z*, *y*, a genes will not be translated
(c) RNA polymerase will bind the promoter region
(d) Only *z* gene will get transcribed

**177.** Tegmina in cockroach, arises from
(a) mesothorax (b) metathorax
(c) prothorax and mesothorax (d) prothorax

**178.** Which of the following is not the function of conducting part of respiratory system?
(a) Inhaled air is humidified
(b) Temperature of inhaled air is brought to body temperature
(c) Provides surface for diffusion of $O_2$ and $CO_2$
(d) It clears inhaled air from foreign particles

**179.** Which of the following is a correct match for disease and its symptoms?
(a) Tetany-High $Ca^{2+}$ level causing rapid spasms
(b) Myasthenia gravis–Genetic disorder resulting in weakening and paralysis of skeletal muscle
(c) Muscular dystrophy–An autoimmune disorder causing progressive degeneration of skeletal muscle
(d) Arthritis–Inflammed joints

**180.** Regarding meiosis, which of the statements is incorrect?
(a) DNA replication occurs in S-phase of meiosis-II
(b) Pairing of homologous chromosomes and recombination occurs in meiosis-I
(c) Four haploid cells are formed at the end of meiosis-II
(d) There are two stages in meiosis, meiosis-I and II

**181.** Detritivores breakdown detritus into smaller particles. This process is called
(a) fragmentation (b) humification
(c) decomposition (d) catabolism

**182.** If the length of a DNA molecule is 1.1 metres, what will be the approximate number of base pairs?
(a) $6.6 \times 10^9$ bp (b) $3.3 \times 10^6$ bp
(c) $6.6 \times 10^6$ bp (d) $3.3 \times 10^9$ bp

***183.** Which of the following functions is not performed by secretions from salivary glands?
(a) Digestion of complex carbohydrates
(b) Lubrication of oral cavity
(c) Digestion of disaccharides
(d) Control bacterial population in mouth

**184.** A dehydration reaction links two glucose molecules to produce maltose. If the formula for glucose is $C_6H_{12}O_6$ then what is the formula for maltose?
(a) $C_{12}H_{24}O_{12}$ (b) $C_{12}H_{22}O_{11}$
(c) $C_{12}H_{24}O_{11}$ (d) $C_{12}H_{20}O_{10}$

**185.** If '8' *Drosophila* in a laboratory population of '80' died during a week, the death rate in the population is ..... individuals per *Drosophila* per week.
(a) 10 (b) 1.0 (c) zero (d) 0.1

## Section B

***186.** Given below are two statements.

**Statement I** In a scrubber, the exhaust from the thermal plant is passed through the electric wires to charge the dust particles.

**Statement II** Particulate matter (Pm 2.5) cannot be removed by scrubber but can be removed by an electrostatic precipitator.

In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both statement I and statement II are incorrect
(b) Statement I is correct, but statement II is incorrect
(c) Statement I is incorrect, but statement II is correct
(d) Both statement I and statement II are correct

**187.** Which of the following is a correct statement?
(a) Bacteria are exclusively heterotrophic organisms
(b) Slime moulds are saprophytic organisms classified under kingdom–Monera
(c) *Mycoplasma* have DNA, Ribosome and cell wall
(d) Cyanobacteria are a group of autotrophic organisms classified under kingdom–Monera

**188.** Ten *E.coli* cells with $^{15}N$-*ds*DNA are incubated in medium containing $^{14}N$ nucleotide. After 60 minutes, how many *E.coli* cells will have DNA totally free from $^{15}N$?
(a) 40 cells (b) 60 cells (c) 80 cells (d) 20 cells

**189.** Select the incorrect statement with respect to acquired immunity.
(a) Anamnestic response is elicited on subsequent encounters with the same pathogen
(b) Anamnestic response is due to memory of first encounter
(c) Acquired immunity is non-specific type of defence present at the time of birth
(d) Primary response is produced when our body encounters a pathogen for the first time

**190.** Statements related to human insulin are given below. Which statement(s) is/are correct about genetically engineered insulin?
I. Pro-hormone insulin contains extra stretch of C-peptide.
II. A-peptide and B-peptide chains of insulin were produced separately in *E.coli*, extracted and combined by creating disulphide bond between them.

III. Insulin used for treating diabetes was extracted from cattles and pigs.
IV. Pro-hormone insulin needs to be processed for converting into a mature and functional hormone.
V. Some patients develop allergic reactions to the foreign insulin.

Choose the most appropriate answer from the options given below.
(a) Only II (b) III and IV
(c) III, IV and V (d) I, II and IV

**191.** Which of the following are not the effects of parathyroid hormone?
I. Stimulates the process of bone resorption.
II. Decreases $Ca^{2+}$ level in blood.
III. Reabsorption of $Ca^{2+}$ by renal tubules.
IV. Decreases the absorption of $Ca^{2+}$ from digested food.
V. Increases metabolism of carbohydrates.

Choose the most appropriate answer from the options given below.
(a) II, IV and V (b) I and V (c) II and III (d) I and III

**192.** Select the incorrect statement regarding synapses.
(a) Electrical current can flow directly from one neuron into the other across the electrical synapse
(b) Chemical synapses use neurotransmitters
(c) Impulse transmission across a chemical synapse is always faster than that across an electrical synapse
(d) The membranes of pre-synaptic and post-synaptic neurons are in close proximity in an electrical synapse

**193.** Match List-I with List-II.

| | List-I | | List-II |
|---|---|---|---|
| A. | Bronchioles | (i) | Dense regular connective tissue |
| B. | Goblet cell | (ii) | Loose connective tissue |
| C. | Tendons | (iii) | Glandular tissue |
| D. | Adipose tissue | (iv) | Ciliated epithelium |

Choose the correct answer from the options given below.
(a) (A) - (i), (B) - (ii), (C) - (iii), (D) - (iv)
(b) (A) - (ii), (B) - (i), (C) - (iv), (D) - (iii)
(c) (A) - (iii), (B) - (iv), (C) - (ii), (D) - (i)
(d) (A) - (iv), (B) - (iii), (C) - (i), (D) - (ii)

**194.** The recombination frequency between the genes *a* and *c* is 5%, *b* and *c* is 15%, *b* and *d* is 9%, *a* and *b* is 20%, *c* and *d* is 24% and *a* and *d* is 29%. What will be the sequence of these genes on a linear chromosome?
(a) *d, b,a, c* (b) *a, b, c, d*
(c) *a, c, b, d* (d) *a, d, b, c*

**195.** If a colourblind female marries a man whose mother was also colourblind, what are the chances of her progeny having colour blindness?
(a) 50% (b) 75%
(c) 100% (d) 25%

**196.** Match List-I with List-II.

| | List-I (Biological molecules) | | List - II (Biological functions) |
|---|---|---|---|
| A. | Glycogen | (i) | Hormone |
| B. | Globulin | (ii) | Biocatalyst |
| C. | Steroids | (iii) | Antibody |
| D. | Thrombin | (iv) | Storage product |

Choose the correct answer from the options given below.
(a) (A) - (iv), (B) - (ii), (C) - (i), (D) - (iii)
(b) (A) - (ii), (B) - (iv), (C) - (iii), (D) - (i)
(c) (A) - (iv), (B) - (iii), (C) - (i), (D) - (ii)
(d) (A) - (iii), (B) - (ii), (C) - (iv), (D) - (i)

**197.** Which of the following is not a desirable feature of a cloning vector?
(a) Presence of a marker gene
(b) Presence of single restriction enzyme site
(c) Presence of two or more recognition sites
(d) Presence of origin of recognition sites

**198.** Which of the following statements is not true?
(a) Sweet potato and potato is an example of analogy
(b) Homology indicates common ancestry
(c) Flippers of penguins and dolphins are a pair of homologous organs
(d) Analogous structures are a result of convergent evolution

**199.** Which one of the following statements is correct?
(a) The tricuspid and the bicuspid valves open due to the pressure exerted by the simultaneous contraction of the atria
(b) Blood moves freely from atrium to the ventricle during joint diastole
(c) Increased ventricular pressure causes closing of the semilunar valves
(d) The Atrio-Ventricular Node (AVN) generates an action potential to stimulate atrial contraction

**200.** Match List-I with List-II with respect to methods of contraception and their respective actions.

| | List-I | | List-II |
|---|---|---|---|
| A. | Diaphragms | (i) | Inhibit ovulation and implantation |
| B. | Contraceptive pills | (ii) | Increase phagocytosis of sperm within uterus |
| C. | Intra Uterine Devices | (iii) | Absence of menstrual cycle and ovulation following parturition |
| D. | Lactational amenorrhea | (iv) | They cover the cervix blocking the entry of sperms |

Choose the correct answer from the options given below.
(a) (A) - (iv), (B) - (i), (C) - (ii), (D) - (iii)
(b) (A) - (ii), (B) - (iv), (C) - (i), (D) - (iii)
(c) (A) - (iii), (B) - (ii), (C) - (i), (D) - (iv)
(d) (A) - (iv), (B) - (i), (C) - (iii), (D) - (ii)

# Answers

## Physics

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1. (d) | 2. (c) | 3. (a) | 4. (b) | 5. (b) | 6. (b) | 7. (a) | 8. (b) | 9. (b) | 10. (b) |
| 11. (c) | 12. (a) | 13. (c) | 14. (d) | 15. (b) | 16. (a) | 17. (b) | 18. (b) | 19. (b) | 20. (a) |
| 21. (a) | 22. (d) | 23. (c) | 24. (c) | 25. (*) | 26. (a) | 27. (a) | 28. (c) | 29. (a) | 30. (a) |
| 31. (b) | 32. (b) | 33. (c) | 34. (b) | 35. (c) | 36. (b) | 37. (a) | 38. (b) | 39. (a) | 40. (b) |
| 41. (c) | 42. (*) | 43. (c) | 44. (a) | 45. (a) | 46. (b) | 47. (a) | 48. (b) | 49. (b) | 50. (d) |

## Chemistry

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 51. (b) | 52. (c) | 53. (d) | 54. (c) | 55. (a) | 56. (c) | 57. (a) | 58. (b) | 59. (b) | 60. (a) |
| 61. (d) | 62. (c) | 63. (d) | 64. (a) | 65. (d) | 66. (c) | 67. (a) | 68. (a) | 69. (b) | 70. (c) |
| 71. (b) | 72. (b) | 73. (d) | 74. (c) | 75. (a) | 76. (b) | 77. (c) | 78. (a) | 79. (b) | 80. (c) |
| 81. (d) | 82. (d) | 83. (d) | 84. (c) | 85. (c) | 86. (b) | 87. (c) | 88. (a) | 89. (*) | 90. (c) |
| 91. (d) | 92. (d) | 93. (b) | 94. (d) | 95. (a) | 96. (d) | 97. (d) | 98. (d) | 99. (a) | 100. (a) |

## Botany

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 101. (a) | 102. (d) | 103. (b) | 104. (d) | 105. (c) | 106. (a) | 107. (b) | 108. (a) | 109. (c) | 110. (d) |
| 111. (b) | 112. (c) | 113. (d) | 114. (a) | 115. (b) | 116. (d) | 117. (b) | 118. (d) | 119. (b) | 120. (a) |
| 121. (a) | 122. (b) | 123. (b) | 124. (b) | 125. (b) | 126. (b) | 127. (b) | 128. (a) | 129. (c) | 130. (b) |
| 131. (b) | 132. (a) | 133. (a) | 134. (a) | 135. (a) | 136. (a) | 137. (b) | 138. (a) | 139. (a) | 140. (a) |
| 141. (d) | 142. (a) | 143. (a) | 144. (a) | 145. (b) | 146. (a) | 147. (d) | 148. (b) | 149. (b) | 150. (d) |

## Zoology

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 151. (b) | 152. (c) | 153. (a) | 154. (b) | 155. (a) | 156. (d) | 157. (a) | 158. (b) | 159. (a) | 160. (c) |
| 161. (b) | 162. (c) | 163. (d) | 164. (c) | 165. (b) | 166. (c) | 167. (a) | 168. (d) | 169. (d) | 170. (b) |
| 171. (c) | 172. (b) | 173. (a) | 174. (d) | 175. (d) | 176. (b) | 177. (a) | 178. (c) | 179. (d) | 180. (a) |
| 181. (a) | 182. (d) | 183. (c) | 184. (b) | 185. (d) | 186. (c) | 187. (d) | 188. (b) | 189. (c) | 190. (a) |
| 191. (a) | 192. (c) | 193. (d) | 194. (c) | 195. (c) | 196. (c) | 197. (c) | 198. (c) | 199. (b) | 200. (a) |

***Note*** *(*) None of the option is correct.*

# Answer with Explanations

## Physics

**1.** *(d)* We know that pressure inside the soap bubble of radius $r$ is given as,

$$p = \frac{4T}{r}$$

where, $T$ is surface tension.

$$\Rightarrow \qquad p \propto \frac{1}{r} \qquad \ldots (i)$$

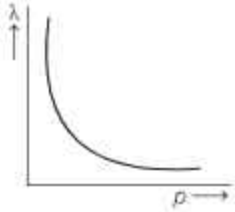
Therefore, when soap bubble expands, then its radius increases, hence from Eq. (i), pressure inside it will decrease.

**2.** *(c)* de-Broglie wavelength ($\lambda$) of a moving particle is given as

$$\lambda = \frac{h}{p}$$

$$\therefore \qquad \lambda \propto \frac{1}{p}$$

Hence, the graph showing the variation of de-Broglie wavelength ($\lambda$) of a moving particle with its momentum ($p$) is shown below.




**3.** *(a)* Given, mass of body, $m = 60$ g

$$= 6 \times 10^{-2} \text{ kg}$$

Gravitational force, $F = 3 \cdot 0$ N

Magnitude of gravitational field intensity

$$I = \frac{F}{m} = \frac{3}{6 \times 10^{-2}} = 50 \text{ N/kg}$$

**4.** *(b)* According to Biot-Savart's law, magnetic field due to small current element $Id\mathbf{l}$, at a distance $r$ from it is given as

$$dB = \frac{\mu_0 (Id\mathbf{l} \times \mathbf{r})}{4\pi r^3}$$

Thus, current element $Id\mathbf{l}$ is a vector quantity. Hence, statement-I is correct and statement-II is incorrect.

**5.** *(b)* Initial speed of freely falling body,

$$u = 0$$

We know that, distance travelled by freely falling body in $n$th second

$$s_n = u + \frac{1}{2}g(2n-1) = 0 + \frac{1}{2}g(2n-1)$$

$$s_n = \frac{1}{2}g(2n-1)$$

Distance travelled by body in first second ($n = 1$ s)

$$s_1 = \frac{1}{2}g(2 \times 1 - 1) = \frac{g}{2}$$

Distance travelled by body in 2nd second ($n = 2$ s)

$$s_2 = \frac{1}{2}g(2 \times 2 - 1) = \frac{3g}{2}$$

Distance travelled by body in 3rd second ($n = 3$ s)

$$s_3 = \frac{1}{2}g\,(2 \times 3 - 1) = \frac{5g}{2}$$

Distance travelled by body in 4th second ($n = 4$ s)

$$s_4 = \frac{1}{2}g(2 \times 4 - 1) = \frac{7g}{2}$$

$$\therefore \; s_1 : s_2 : s_3 : s_4 = \frac{g}{2} : \frac{3g}{2} : \frac{5g}{2} : \frac{7g}{2}$$

$$= 1 : 3 : 5 : 7$$

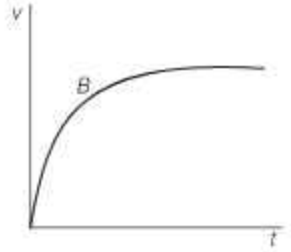
**6.** *(b)* In half wave rectification, input frequency, $f_{in} = 60$ Hz

Since, frequency of output voltage obtained from the half wave rectifier is same as frequency of input AC voltage.

Hence, $f_{out} = 60$ Hz

**7.** *(a)* When a spherical ball is dropped in a long column of a highly viscous liquid, then its velocity first increases and after sometime, it moves with constant velocity, known as terminal velocity ($v$).

Hence, correct variation of speed of ball in liquid column with time is given as




**8.** *(b)* Work done to move a charge ($q$) particle between any two points on equipotential surface is zero.

i.e. $\qquad W = q(V_2 - V_1)$

$$\Rightarrow \qquad \mathbf{F} \cdot d\mathbf{s} = q(V - V) \quad [\because V_1 = V_2 = V]$$

$$\Rightarrow \qquad q(\mathbf{E} \cdot d\mathbf{s}) = 0 \quad [\because \mathbf{F} = \mathbf{E} \cdot d\mathbf{s}]$$

$$\Rightarrow \qquad \mathbf{E} \cdot d\mathbf{s} = 0$$

Therefore, $\mathbf{E} \perp d\mathbf{s}$

i.e., electric field is perpendicular to displacement vector ($d\mathbf{s}$).

Since, $d\mathbf{s}$ lies in the plane of equipotential surface.

Thus, angle between the electric lines of force and the equipotential surface is 90°.

**9.** *(b)* Coefficient of thermal resistance $\alpha$ is positive for conductor and negative for semiconductor. Thus, on increasing temperature, resistance of conductor increases and resistance of semiconductor decreases.

**10.** *(b)* The given nuclear reaction is represented as

$$^{22}_{11}\text{Na} \longrightarrow {}^{M}_{Z}X + e^{+} + \nu$$

Since, atomic number and mass number are same for both side of reaction.

Hence, for atomic number,

$$11 = Z + 1$$

$$\Rightarrow \qquad Z = 10$$

For mass number,

$$22 = M \Rightarrow M = 22$$

$$\therefore \qquad {}^{M}_{Z}X = {}^{22}_{10}\text{Ne}$$

**11.** *(c)* For the first excited state of hydrogen atom, $n_1 = 2$

For the second excited state of hydrogen atom,

$$n_2 = 3$$

According to Bohr's model of H-atom, energy of an electron revolving in $n$th orbit.

$$E = \frac{-13 \cdot 6}{n^2} \text{ eV}$$

Here, for the first excited state.

$$T_1 = E_1 = \frac{-13 \cdot 6}{n_1^2} = \frac{-13 \cdot 6}{2^2} \qquad [\because n_1 = 2]$$

For the second excited state,

$$T_2 = E_2 = \frac{-13 \cdot 6}{n_2^2} = \frac{-13 \cdot 6}{3^2} \text{ eV}$$

$$[\because n_2 = 3]$$

$$\therefore \; \frac{T_1}{T_2} = \frac{-13 \cdot 6 / 2^2}{-13 \cdot 6 / 3^2} = \frac{3^2}{2^2} = \frac{9}{4}$$

$$\Rightarrow T_1 : T_2 = 9 : 4$$

**12.** *(a)* Radius of gyration of a body is given as

$$K = \sqrt{\frac{I}{m}}$$

Where, $I$ is moment of inertia and $m$ is mass of body.

$$\therefore \quad \frac{K_1}{K_2} = \sqrt{\frac{I_1}{I_2}} \qquad \ldots (i)$$

$I_1$ = Moment of inertia of thin uniform disc about an axis passing through its centre and normal to plane.

$$= \frac{MR^2}{2} \quad \begin{pmatrix} \text{where, } R = \text{radius of disc} \\ \text{and} \quad M = \text{mass of disc} \end{pmatrix}$$

$I_2$ = Moment of inertia of disc about its diameter.

$$= \frac{MR^2}{4}$$

$\therefore$ From Eq. (i),

$$\frac{K_1}{K_2} = \sqrt{\frac{MR^2/2}{MR^2/4}} = \sqrt{2}$$

$$\therefore \quad K_1 : K_2 = \sqrt{2} : 1$$

**13.** *(c)* When light (EM wave) propagates through a material medium of relative permittivity $\varepsilon_r$ and relative permeability $\mu_r$, then velocity of light (EM wave) is given by

$$v = c / \sqrt{\mu_r \varepsilon_r}$$

where, $c$ is speed of light in vacuum.

**14.** *(d)* Given, power radiated by transmitter,

$$P = 100 \text{ kW} = 10^5 \text{ W}$$

$\therefore$ Time, $t = 1 \text{ h} = 60 \times 60 \text{ s} = 3{\cdot}6 \times 10^3 \text{ s}$

$\therefore$ Energy radiated by the transmitter,

$$E = P \times t$$
$$= 10^5 \times 3{\cdot}6 \times 10^3$$
$$= 3{\cdot}6 \times 10^8 \text{ J} = 36 \times 10^7 \text{ J}$$

**15.** *(b)* Given, mass (lift + passengers),

$$m = 2000 \text{ kg}$$

$\therefore$ Weight (lift + passengers),

$$w = mg = 2000 \text{ g N}$$
$$v = 1{\cdot}5 \text{ m/s}$$

Frictional force, $f = 3000$ N

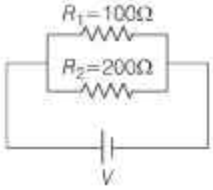
Since, lift is moving with constant velocity, hence net applied force on the lift = friction force + weight.

$$F = 3000 + 2000 \text{ g}$$
$$= 3000 + 2000 \times 10 = 23000 \text{ N}$$

$\therefore$ Minimum power delivered, $P = F \times v$

$$= 23000 \times 1.5 = 34500 \text{ W}.$$

**16.** *(a)* The given situation is shown in the following circuit diagram.




Let, these two resistors are connected with a battery of voltage $V$ volts. We know that, in parallel combination, potential difference across each resistor is same.

Hence, potential difference across $R_1$, and $R$ are given as,

$$V_1 = V_2 = V \text{ volts}$$

If $H_1$ and $H_2$ are thermal energy produced through resistances $R_1$ and $R_2$ in time $t$, then

$$H_1 = \frac{V_1^2}{R_1} t = \frac{V^2 t}{100}$$

and

$$H_2 = \frac{V_2^2 t}{R_2} = \frac{V^2 t}{200}$$

Hence, $\frac{H_1}{H_2} = \frac{V^2 t/100}{V^2 t/200} = \frac{200}{100} = \frac{2}{1}$

$$\Rightarrow \quad H_1 : H_2 = 2 : 1$$

**17.** *(b)* We know that, peak voltage ($V_0$) and rms voltage ($V_{rms}$) are related as,

$$V_0 = \sqrt{2}\, V_{rms}$$

Thus, peak voltage of the AC source is equal to $\sqrt{2}$ times the rms value of the AC source.

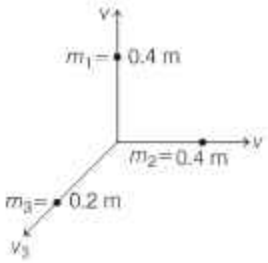
**18.** *(b)* Let total mass of the shell be $m$.

$\therefore$ The ratio of the masses of the fragment is 2 : 2 : 1

$$m_1 + m_2 + m_3 = m$$
$$2k + 2k + k = m$$
$$\Rightarrow \quad k = 0.2 \text{ m}$$

$\therefore$ Mass of lighter fragment,

$$m_3 = k = 0.2 \text{ m}$$
$$m_1 = m_2 = 2 \times 0.2 \text{ m} = 0.4 \text{ m}$$




$\therefore$ Let the velocity of lighter fragment is $v_3$,

Since, $p_3 = \sqrt{p_1^2 + p_2^2}$

$$\Rightarrow \quad 0{\cdot}2mv_3 = \sqrt{(0{\cdot}4mv)^2 + (0{\cdot}4mv)^2}$$
$$\Rightarrow \quad v_3 = 2\sqrt{2}\, v$$

**19.** *(b)* We know that, velocity of transverse wave in the string, $v \propto \sqrt{T}$

Where, $T$ is tension in the string.

$$\therefore \quad \frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}} = \sqrt{\frac{T_1}{2T_1}} \qquad [\because T_2 = 2T_1]$$
$$= \frac{1}{\sqrt{2}}$$
$$v_1 : v_2 = 1 : \sqrt{2}$$

**20.** *(a)* When a thermodynamic system undergoes a change in such a way that no exchange of heat takes place between it and the surrounding, the process is known as adiabatic process.

Curve 4 is isobaric process because pressure is constant.

Curve 1 is isochoric process because volume is constant.

In curve 2 and 3, curve 3 has smaller slope than curve 2. Hence, curve 3 represents isothermal process and curve 2 represents adiabatic process.

**21.** *(a)* Given, initial angular velocity of fly wheel,

$$\omega_0 = 2\pi \text{ rad} \times 1200 \text{ rpm}$$
$$= 2\pi \text{ rad} \times \frac{1200}{60} \text{ rps}$$
$$= 40\,\pi \text{ rad/s}$$

Similarly, final angular velocity of fly wheel,

$$\omega = 2\pi \text{ rad} \times 3120 \text{ rpm}$$
$$= 2\pi \text{ rad} \times \frac{3120}{60} \text{ rps}$$
$$= 104\,\pi \text{ rad/s}$$

Time, $t = 16$ s

If $\alpha$ be the angular acceleration, then using the equation of rotational motion,

$$\omega = \omega_0 + \alpha t$$
$$\Rightarrow \quad 104\pi = 40\pi + \alpha \times 16$$
$$\Rightarrow \quad \alpha = \frac{104\,\pi - 40\,\pi}{16}$$
$$= \frac{64\,\pi}{16} = 4\pi \text{ rad/s}^2$$

**22.** *(d)* Plane angle is measured in radian whereas solid angle is measured in steradian.

Therefore, plane angle and solid angle have units but no dimensions.

**23.** *(c)* In the circuit diagram (A) and (C), both the $p$-$n$ junction diodes are in same biasing conditions.

So, both diodes offers equal resistances.

Since, both diodes are in series, therefore equal potential will drop across the junction.

**24.** *(c)* In Young's double slit experiment, wavelength of light used is inversely proportional to the number of fringes observed.

i.e. $\lambda_1 n_1 = \lambda_2 n_2$

$$n_2 = \frac{n_1\lambda_1}{\lambda_2}$$

Here, $\lambda_1 = 600\text{ nm} = 6\times10^{-7}\text{ m}$

$\lambda_2 = 400\text{ nm} = 4\times10^{-7}\text{ m}$

$n_1 = 8$

$$\therefore \quad n_2 = \frac{8\times6\times10^{-7}}{4\times10^{-7}} = 12$$

**25.** *(*)* Using Einstein's photoelectric equation,

$$eV' = h\nu' - h\nu_0$$

when $\nu' = \nu$ and $V' = V_s$, then

$$\therefore \quad eV_s = h\nu - h\nu_0 \qquad \ldots (i)$$

Again when $\nu' = \frac{\nu}{2}$ and $V' = \frac{V_s}{2}$

$$\therefore \quad \frac{eV_s}{2} = \frac{h\nu}{2} - h\nu_0 \qquad \ldots (ii)$$

Dividing Eq. (i) by Eq. (ii), we have

$$\frac{eV_s}{eV_s/2} = \frac{h\nu - h\nu_0}{\frac{h\nu}{2} - h\nu_0}$$

$$2 = \frac{h(\nu-\nu_0)}{h\left(\frac{\nu}{2}-\nu_0\right)}$$

$$\Rightarrow 2\left(\frac{\nu}{2}-\nu_0\right) = \nu - \nu_0$$

$$\nu - 2\nu_0 = \nu - \nu_0 \Rightarrow \nu_0 = 0$$

Hence, question is wrong.

**26.** *(a)* Side of square loop, $a = 1$ m

∴Area of square loop, $A = a^2 = 1^2 = 1\text{ m}^2$

Resistance of square loop, $R = 1\,\Omega$

Magnetic field, $B = 0{\cdot}5$ T

Since, plane of loop is perpendicular to the direction of magnetic field, hence angle made by the normal to the loop with the direction of magnetic field will be $0^\circ$.

i.e., $\theta = 0$

∴Magnetic flux passing through the loop,

$$\phi = BA\cos\theta = 0.5\times1\times\cos0^\circ$$
$$= 0.5\text{ Wb}$$

**27.** *(a)* Potential on the hollow conducting sphere of radius $R$ due to charge $Q$ is

$$V = \frac{1}{4\pi\varepsilon_0}\cdot\frac{Q}{R}$$

$$\Rightarrow \quad V \propto \frac{1}{R} \quad [\text{For same charge } Q]$$

$$\Rightarrow \quad \frac{V_1}{V_2} = \frac{R_2}{R_1}$$

Since, $R_1 > R_2$

$$\therefore \quad \frac{R_1}{R_2} > 1 \Rightarrow \frac{R_2}{R_1} < 1$$

$$\therefore \quad \frac{V_1}{V_2} < 1$$

$$\therefore \quad V_1 < V_2$$

Hence, potential will more on smaller sphere.

**28.** *(c)* Given, radius, $r = \frac{10^{-2}}{\sqrt{\pi}}$ m

$\therefore R = 10\,\Omega \Rightarrow E = 10$ V / m

$l = 10$ m

∴ Area of cross-section of copper wire,

$$A = \pi r^2$$

$$= \pi\times\left(\frac{10^{-2}}{\sqrt{\pi}}\right)^2$$

$$= 10^{-4}\text{ m}^2$$

∴Current density,

$$J = \frac{i}{A} = \frac{V/R}{A} \qquad \left[\because i = \frac{V}{R}\right]$$

$$= \frac{V}{RA} = \frac{El}{RA} \qquad [\because V = El]$$

$$= \frac{10\times10}{10\times10^{-4}} = 10^5\text{ A / m}^2$$

**29.** *(a)* Given, Radius of solenoid, $r = 1$ mm

$$\text{Number of turns per mm} = \frac{100}{1\text{ mm}}$$

∴Number of turns per metre

$$= \frac{100}{1\times10^{-3}\text{m}}$$

i.e., $n = 10^5$ turns/m

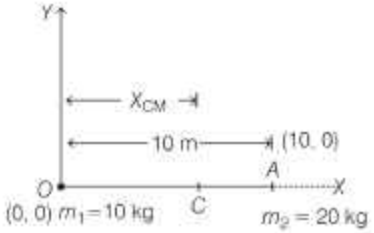
Current flowing through the solenoid,

$$I = 1\text{ A}$$

∴Magnetic field strength at the centre of the solenoid

$$B = \mu_0 nI$$
$$= 4\pi\times10^{-7}\times10^5\times1$$
$$= 4\pi\times10^{-2}\text{ T}$$
$$= 12{\cdot}56\times10^{-2}\text{ T}$$

**30.** *(a)* The given situation is shown below.




Let $X_{CM}$ be the position of centre of mass from 10 kg mass.

$$\therefore \quad X_{CM} = \frac{m_1x_1 + m_2x_2}{m_1 + m_2}$$

$$= \frac{10\times0 + 20\times10}{10+20}$$

$$= \frac{200}{30} = \frac{20}{3}\text{ m}$$

**31.** *(b)* Given, refractive index of glass surface

$$\mu = \sqrt{3}$$

Angle of incidence, $i = 60^\circ$

Since, above situation follows Brewster's law because,

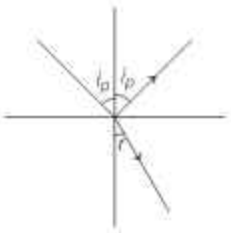
$$\mu = \tan i = \tan60^\circ = \sqrt{3}$$

Hence, angle $i$ is equal to polarising angle $i_p$.

i.e., $i_p = i = 60^\circ$

By snell's law

$$\mu = \frac{\sin i_p}{\sin r}$$




$$\Rightarrow \quad \tan i_p = \frac{\sin i_p}{\sin r} \quad [\text{Since, } \mu = \tan i_p]$$

$$\Rightarrow \quad \frac{\sin i_p}{\cos i_p} = \frac{\sin i_p}{\sin r}$$

$$\Rightarrow \quad \cos i_p = \sin r$$

$\Rightarrow \quad \cos i_p = \cos(90° - r)$

$\Rightarrow \quad i_p = 90° - r$

$\Rightarrow \quad r + i_p = 90°$

Hence, angle between the refracted and reflected rays

$= 180° - (r + i_p) = 180° - (90°) = 90°$

It is also called Brewster's law.

**32.** *(b)* We know that, magnetic field due to current carrying solenoid,

$$B = \mu_0 nI$$

$$\Rightarrow \quad \mu_0 = \frac{B}{nI}$$

$$\Rightarrow [\mu_0] = \frac{[B]}{[n][I]} = \frac{\left[\frac{F}{IL}\right]}{[n][I]} \quad \left[\because F = IBL \Rightarrow B = \frac{F}{IL}\right]$$

$$= \frac{[F]}{[n][L][I^2]} = \frac{[MLT^{-2}]}{[L^{-1}][L][A^2]}$$

$$= [MLT^{-2}A^{-2}]$$

Hence, option (b) is correct.

Dimensional formula of self inductance, $[L] = [M^1L^2T^{-2}A^{-2}]$

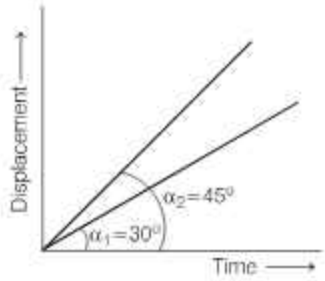
Dimensional formula of electric permittivity,

$$[\varepsilon_0] = [M^{-1}L^{-3}T^4A^2]$$

Dimensional formula of magnetic flux,

$$[\phi] = [ML^2T^{-2}A^{-1}]$$

**33.** *(c)* We know that, slope of displacement-time graph for a moving object gives velocity of moving object.




$\therefore$ Velocity of first moving particle,

$$v_1 = \tan\alpha_1 = \tan 30° = \frac{1}{\sqrt{3}}$$

Velocity of second moving particle,

$$v_2 = \tan\alpha_2 = \tan 45° = 1$$

$$\therefore \quad \frac{v_1}{v_2} = \frac{1/\sqrt{3}}{1} = \frac{1}{\sqrt{3}}$$

$\therefore \quad v_1 : v_2 = 1 : \sqrt{3}$

**34.** *(b)* For a biconvex lens,

$R_1 = +20$ cm, $R_2 = -20$ cm and $\mu = 1.5$

By using lens maker's formula, we have,

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$= (1.5 - 1)\left(\frac{1}{20} - \frac{1}{-20}\right) = 0.5\left(\frac{2}{20}\right)$$

$$\Rightarrow \quad \frac{1}{f} = 0.5 \times \frac{1}{10} \Rightarrow f = \frac{10}{0.5}$$

$$\Rightarrow \quad f = 20 \text{ cm} = 0.2 \text{ m}$$

$\therefore$ Power of the lens, $P = \frac{1}{f}$ (in metre)

$$= \frac{1}{0.2} = +5 \text{ D}$$

**35.** *(c)* A. Wavelength range of AM radio waves lies between $10^2$ m to $10^3$ m.

B. Wavelength range of microwaves lies between $10^{-3}$ m to $10^{-1}$ m.

C. Wavelength range of infrared radiations lies between $10^{-5}$ m to $10^{-3}$ m.

D. Wavelength range of X-rays lies between $10^{-12}$ m to $10^{-8}$ m.

Therefore, correct matching pairs are

(A)-(ii), (B)-(iii), (C)-(iv), (D)-(i)

**36.** *(b)* Given, mass number of parent nucleus, $A = 189$

Mass number of daughters nuclei,

$A_1 = 125$ and $A_2 = 64$

We know that, $R = R_0(A^{1/3})$

where, $R_0$ is constant.

$$\therefore \quad \frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{1/3} = \left(\frac{125}{64}\right)^{1/3}$$

$$= \left[\left(\frac{5}{4}\right)^3\right]^{1/3} = \frac{5}{4}$$

$\therefore \quad R_1 : R_2 = 5 : 4$

**37.** *(a)* Given, speed of light in transparent medium $A$,

$$v_A = 1.5 \times 10^8 \text{ m/s}$$

Speed of light in transparent medium $B$,

$$v_B = 2 \times 10^8 \text{ m/s}$$

Refractive index ($\mu$) and critical angle ($i_c$) are related as

$$\mu = \frac{1}{\sin i_c} \Rightarrow \sin i_c = \frac{1}{\mu}$$

$$= \frac{1}{(v_B / v_A)} \quad \left[\because \mu = \frac{v_B}{v_A}\right]$$

$$= \frac{v_A}{v_B} = \frac{1.5 \times 10^8}{2 \times 10^8} = 0.750$$

$$\Rightarrow \quad i_c = \sin^{-1}(0.750)$$

**38.** *(b)* Given, number of turns in big circular coil, $N = 1000$

Average radius, $r = 10$ m

Vertical component of Earth's magnetic field, $B_V = 2 \times 10^{-5}$ T

Resistance of coil, $R = 12.56\,\Omega$

Angular velocity, $\omega = 2$ rad/s

Maximum induced emf in the coil

$$e_{max} = NBA\omega$$

$$= NB_vA\omega$$

$$= 1000 \times 2 \times 10^{-5} \times \pi(10)^2 \times 2 \quad [\because A = \pi r^2]$$

$$e_{max} = 4\pi \text{ V}$$

$\therefore$ Maximum induced current in the coil

$$I_{max} = \frac{e_{max}}{R} = \frac{4\pi}{12.56} = 1 \text{ A}$$

**39.** *(a)* A. According to Newton's law of gravitation,

$$F = G\frac{m_1m_2}{r^2}$$

$$\Rightarrow \quad G = \frac{Fr^2}{m_1m_2}$$

$$\Rightarrow [G] = \frac{[F][r^2]}{[m_1][m_2]} = \frac{[MLT^{-2}][L^2]}{[M][M]}$$

$$\Rightarrow [G] = [M^{-1}L^3T^{-2}]$$

B. Gravitational potential energy,

$$U = mgh$$

$$[U] = [m][g][h]$$

$$= [M][LT^{-2}][L] = [ML^2T^{-2}]$$

C. Gravitational potential,

$$V = -\frac{Gm}{r}$$

$$\therefore \quad [V] = \frac{[G][m]}{[r]}$$

$$= \frac{[M^{-1}L^3T^{-2}][M]}{[L]}$$

$$= [L^2T^{-2}]$$

D. Gravitational intensity,

$$I = \frac{GM}{r^2}$$

$$\therefore [I] = \frac{[G][M]}{[r^2]}$$

$$= \frac{[M^{-1}L^3T^{-2}][M]}{[L^2]} = [LT^{-2}]$$

Hence, correct matching pair is

(A)-(ii), (B)-(iv), (C)-(i), (D)-(iii)

**40.** *(b)* When a coil spring is stretched, neither its length nor its volume changes, but there is only the change

in its shape. Therefore, stretching of coil spring is determined by shear modulus.

Young's modulus of steel spring is more than Young's modulus of copper spring, therefore tensile strength of steel spring is more than tensile strength of copper spring.

Hence, Assertion is true but Reason is false.

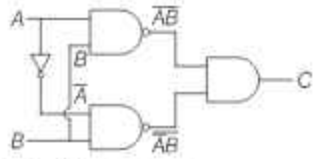
**41.** *(c)* Area = Length × Breadth

$$= 55.3 \times 25 = 1382.5$$

$$= 13.825 \times 10^2 = 13.83 \times 10^2$$

$$= 13.8 \times 10^2 = 14 \times 10^2$$

**42.** *(*)* The given logic circuit is represented as




Output of logic circuit,

$$C = \overline{AB} + \overline{\overline{AB}} = (\overline{A} + \overline{B}) + \overline{\overline{A}} + \overline{B}$$

[Using De-morgan's law $\overline{xy} = \overline{x} + \overline{y}$]

$$= \overline{A} + \overline{B} + A + \overline{B}$$

$$C = \overline{A} + A + \overline{B} = 1 + \overline{B}$$

$[\because \overline{A} + A = 1]$

Truth table is given as

| $A$ | $B$ | $\overline{B}$ | $C = 1 + \overline{B}$ |
|---|---|---|---|
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 |

Hence, no option is correct.

**43.** *(c)* Molecular weight of water = 18 g

$$= 1 \cdot 8 \times 10^{-2} \text{ kg}$$

$\therefore$ Mass of water, $M = 4 \cdot 5$ kg

$\therefore$ number of moles of water,

$$n = \frac{\text{Mass of water}}{\text{Molecular weight of water}}$$

$$= \frac{4.5}{1.8 \times 10^{-2}} = 250$$

At STP, $T = 273$ K, $p = 10^5$ N / m$^2$

Using equation, $pV = nRT$

$$\Rightarrow V = \frac{nRT}{p} = \frac{250 \times 8.3 \times 273}{10^5} = 5.6 \text{ m}^3$$

**44.** *(a)* For the most precise measurement of unknown resistance $X$, the resistances $P$ and $Q$ should be approximately equal and should be small so that it could decrease error in experiment.

**45.** *(a)* Given, In series $L$-$C$-$R$ circuit,

$$L = 10 \text{ H}, C = 10 \mu\text{F} = 10^{-5} \text{ F}$$

$$R = 50 \Omega$$

Source voltage, $V = 200\sin(100\, t)$ volt.

$$\omega = 100 \text{ rad/s}$$

$$\therefore \quad 2\pi\nu = 100$$

$$\Rightarrow \quad \nu = \frac{100}{2\pi} = \frac{50}{\pi} \text{ Hz}$$

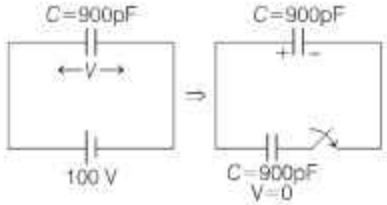
Resonance frequency,

$$\nu_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{10 \times 10^{-5}}}$$

$$= \frac{1}{2\pi \times 10^{-2}} = \frac{50}{\pi} \text{ Hz}$$

Hence, $\nu = \nu_0 = \frac{50}{\pi}$ Hz

**46.** *(b)* The given situation is shown below.




After combination,

Common potential,

$$V = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$$

$$= \frac{C \times 100 + C \times 0}{C + C} = 50 \text{ V}$$

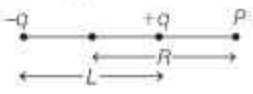
$\therefore$ Electrostatic stored potential energy,

$$U = 2 \times \frac{1}{2} CV^2 = CV^2$$

$$= 9 \times 10^{-10} \times 50^2$$

$$= 2.25 \times 10^{-6} \text{ J}$$

**47.** *(a)* The arrangement of charges shown in the figure is an electric dipole. Hence, electric field at point $P$ at a distance $R (R >> L)$ from the dipole is equal to electric field due to electric dipole on axial position.




i.e., $$E = \frac{1}{4\pi\varepsilon_0} \cdot \frac{2p}{R^3}$$

where, $p = qL$ (Electric dipole moment)

$$\therefore \quad E \propto \frac{1}{R^3}$$

**48.** *(b)* Initial speed of ball, $u = 10 \text{ ms}^{-1}$

Angle of projection, $\theta = 60^\circ$

At highest point of the trajectory of ball, its vertical component of velocity becomes zero and only horizontal component of velocity exists, which is equal to horizontal component of velocity at the time of projection.

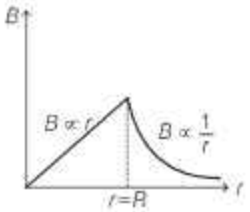
i.e., speed of ball at highest point

$$= u\cos\theta$$

$$= 10\cos 60^\circ = 10 \times \frac{1}{2}$$

$$= 5 \text{ ms}^{-1}$$

**49.** *(b)* According to Ampere's circuital law, magnetic field inside the current carrying long straight wire of circular cross-section is a linearly increasing function of distance $r (r < R)$ upto the boundary of the wire and magnetic field outside the wire at distance $r (r > R)$ from its axis is decreasing one with $\frac{1}{r}$ dependence as shown in figure.




where, $R$ is radius of circular cross-section of wire.

**50.** *(d)* Given, length of first pendulum,

$$l_1 = 121 \text{ cm} = 1 \cdot 21 \text{ m}$$

length of second pendulum,

$$l_2 = 100 \text{ cm} = 1 \text{ m}$$

According to given situation, both pendulums are again in phase if vibrations made by the shorter pendulum will be known.

Let $T_1$ and $T_2$ be the time periods of longer and shorter pendulum respectively.

Since, $T = 2\pi\sqrt{\frac{l}{g}}$

$$\Rightarrow \quad \frac{T_1}{T_2} = \sqrt{\frac{l_1}{l_2}} = \sqrt{\frac{1 \cdot 21}{1}} = 1 \cdot 1 = \frac{11}{10}$$

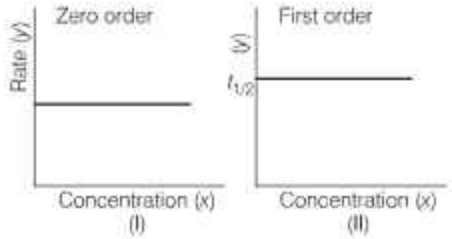
$\Rightarrow 10\,T_1 = 11\,T_2$

$\therefore$ 10 vibrations of longer pendulum = 11 vibrations of shorter pendulum

$\therefore$ Minimum number of vibrations of shorter pendulum = 11

# Chemistry

**51.** *(b)*

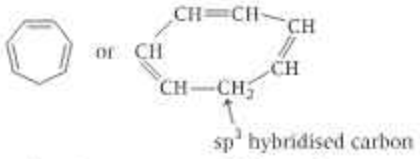



Here, graph I is of zero order if $y$-axis is representing rate and $x$ represents concentration. For zero order, reaction rate is independent of concentration and remains constant.

While II graph represents first order if $y$ is half-life ($t_{1/2}$) and $x$ is concentration. Here $t_{1/2}$ is independent of concentration.

Hence, option (b) is correct, i.e. zero order ($y$ = rate and $x$ = concentration), first order ($y = t_{1/2}$ and $x$ = concentration).

**52.** *(c)*

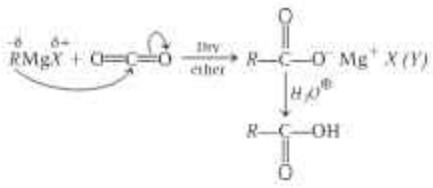



The given structure follows the Huckel's rule $(4n + 2)\pi$ electrons as it has $6\pi$ electrons. But the given structure is non-planar due to the presence of $sp^3$ hybridised carbon atom as shown in the structure. Hence, it is non-aromatic compound. While other three options are planar and follow Huckel's rule. Hence they are aromatic.

So, option (c) is correct.

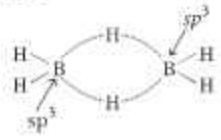
**53.** *(d)* The alkyl group of Grignard reagent have partial negative charge while magnesium will have partial positive charge. The alkyl group will attack carbon atom of $CO_2$ and will form the salts of carboxylic acids, which on hydrolysis will form carboxylic acids.

The reaction takes place is as follows




Hence 'Y' in the reaction is $RCOO^-Mg^+X$.

**54.** *(c)* The incorrect statement about diborane is option (c) because boron has $sp^3$-hybridisation and it is non-planar.




**55.** *(a)* The electronic configuration of Gadolinium is

$$Gd = [Xe]4f^7 5d^1 6s^2.$$

$$Gd^{3+} = [Xe]4f^7$$

After removal of 3rd electron, it will attain half-filled electronic configuration. Due to which $4f$ will have high exchange energy. So, Gd has low value of third ionisation energy.

Hence, option (a) is correct.

**56.** *(c)* According to Dalton's law of partial pressure

Partial pressure of gas = mole fraction of gas in gaseous mixture × total pressure of gaseous mixture.

So, $p_1 = x_1 p$

$p_2 = x_2 p$

$p_3 = x_3 p$

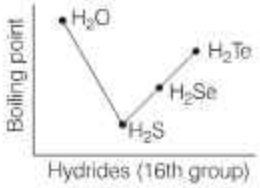
Then, the total pressure is

$p = p_1 + p_2 + p_3$.

Hence, statement (c) is incorrect.

**57.** *(a)* **Statement I** The correct order of boiling points of group 16th hydride is

$H_2S < H_2Se < H_2Te < H_2O$

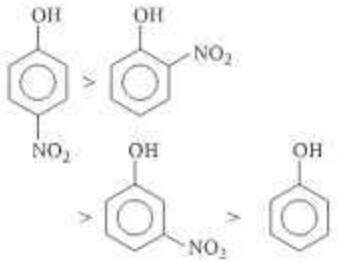



**Statement II** Boiling point increases with increase in molar mass but $H_2O$ has a higher boiling point due to presence of hydrogen bond.

Hence, both statement I and statement II are incorrect.

So, option (a) is correct.

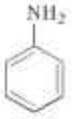
**58.** *(b)* Statement I is correct but statement II is incorrect because, the order of acidic strength is as follows.




—$NO_2$ group at *ortho* and *para*-position withdraws electrons of the O—H bonds towards itself by the stronger $-R$-effect while the $NO_2$ group at *meta* position withdraws electrons of O—H bond by the weaker $-I$-effect. Thus, *ortho* and *para*-nitrophenol are more acidic than *meta*-nitrophenol. Among *ortho* and *para*-nitrophenols, *o*-nitrophenol is less acidic due to intramolecular hydrogen bonding which makes loss of proton a little more difficult.

Hence option (b) is correct.

**59.** *(b)* Kjeldahl's method is not applicable to the compounds containing nitrogen as nitro and azo groups and nitrogen present in the ring (Pyridine). It is because nitrogen of these compounds does not change into ammonium sulphate. Hence in option (a), (c) and (d) Kjeldahl's method is not applicable. It is applicable only in compound (b), where —$NH_2$— group can be converted into ammonium sulphate.




Hence correct option is (b).

**60.** *(a)* The correct match is

A-(iii), B-(iv), C-(ii), D-(i)

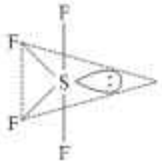
**Explanation**

- **Cimetidine** is an antacid, used to prevent the interaction of histamine with the receptors present in the stomach wall, due to which less amount of acid is released.
- **Seldane** is an antihistamine. It interferes with the natural action of histamine by competing with histamine for binding sites of receptor where histamine exerts its effect.
- **Morphine** is an analgesic. They are chiefly used for the relief of pain, cardiac pain etc.

- **Salvarsan** is an antimicrobial. It mainly affects the bacteria spirochete which causes syphilis in human being.

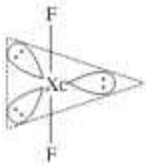
**61.** *(d)* The roots for 1, 1 and 9 are un, un and enn respectively. Hence the symbol and name respectively are uue and ununennium.

**62.** *(c)* **$SF_4$**



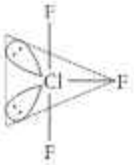
$SF_4$ contain 1 lone-pair

**$XeF_2$**



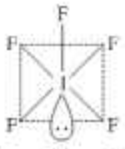
$XeF_2$ contain 3 lone-pairs

**$ClF_3$**



$ClF_3$ contains 2 lone-pairs

**$IF_5$**



Among all the options $XeF_2$ has maximum number of lone pairs, i.e. 3. Hence, it will show maximum 'lone pair- lone pair' repulsion.

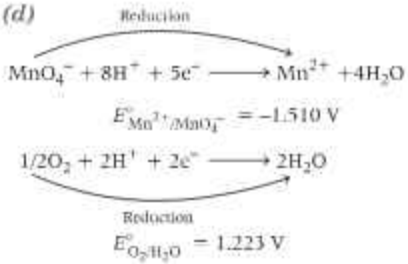
**63.** *(d)* To synthesise chlorobenzene, benzene reacts with $Cl_2$ in the presence of $FeCl_3$. The reaction taking place is as follows




**64.** *(a)* In *p-V* graph, the area under the curve represents the magnitude of work. The maximum area is covered in graph(a).

Hence, it represents maximum work done.

**65.** *(d)*




Reaction at cathode (reduction)

$$2MnO_4^- + 16H^+ + 10e^- \longrightarrow 2Mn^{2+} + 8H_2O$$

$$E^\circ_{RP} = 1.510\ V$$

Reaction at anode (oxidation)

$$5H_2O \longrightarrow \frac{5}{2}O_2 + 10H^+ + 10e^-$$

$$E^\circ_{OP} = -1\cdot223\ V$$

The overall reaction is

$$2MnO_4^- + 6H^+ \longrightarrow 2Mn^{2+} + \frac{5}{2}O_2 + 3H_2O$$

$$E^\circ = E^\circ_{MnO_4^-/Mn^{2+}} - E^\circ_{O_2/H_2O}$$

$$E^\circ = 1.510 - 1.223 = 0.287\ V$$

Here, since the value of $E^\circ$ is positive, the reaction is spontaneous. Hence, $MnO_4^-$ will liberate $O_2$.

**66.** *(c)* The complex $[Ag(H_2O)_2][Ag(CN)_2]$ has coordination number 2 and oxidation state is $Ag^+$ in both cationic and anionic parts.

Hence, its IUPAC name is diaquasilver (I) dicyanidoargentate (I).

**67.** *(a)* The incorrect statement among the given option is option (a).

The oxidation number of K in $KO_2$ is + 1 because in this compound oxygen is present as superoxide ion $K^+O_2^-$ ($O_2^-$ superoxide ion).

**68.** *(a)* $CaCO_3(s) + 2\ HCl(aq) \longrightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$

Number of moles of $CaCO_3$ (pure) = 1 / 2 × number of moles of HCl

[Moles = Molarity × Volume (in L)]

$$= \frac{1}{2} \times 0.5 \times \frac{50}{1000} = 0.0125$$

Weight of $CaCO_3$ (pure)

= Mole × Molecular weight

= 0.0125 × 100 = 1.25 g

% Purity

$$= \frac{\text{Weight of pure substance}}{\text{Weight of impure sample}} \times 100$$

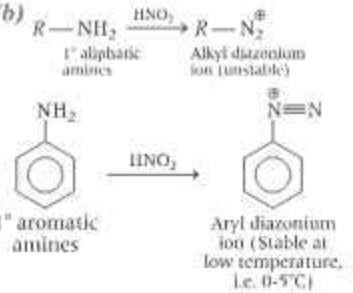
$$95 = \frac{1.25}{\text{Weight of impure sample}} \times 100$$

Weight of impure sample

$$= \frac{1.25 \times 100}{95} = 1.32\ g.$$

Hence, weight of impure sample is 1.32 g.

**69.** *(b)*




Hence, statement I is correct but statement II is incorrect. So, option (b) is correct.

**70.** *(c)* The incorrect statement regarding polymers is option (c). It is because thermosetting polymers are not reusable due to presence of cross linking in them. This cross-linking makes the thermosetting polymer more resistant to high temperature and prevent remelting while all other options are correct.

**71.** *(b)* The incorrect statement regarding chirality is option (b). It is because enantiomers are non-superimposable mirror images of each other.

**72.** *(b)* **Statement I** According to Hardy Schulze rule for negative sol coagulation, greater the charge on cation, greater is its flocculation power. Hence, this statement is correct.

**Statement II** For positive sol coagulation, the greater the charge on anion, greater is its flocculating power. Hence the correct order is $NaCl < Na_2SO_4 < Na_3PO_4$.

So, statement II is incorrect.

Hence option (b) is correct.

**73.** *(d)* The correct answer from the given options is

A-(iv), B-(i), C-(ii), D-(iii)

| Hydride | Nature |
|---|---|
| $MgH_2$ | Ionic |
| $GeH_4$ | Electron precise |
| $B_2H_6$ | Electron deficient |
| HF | Electron rich |

- Electron deficient hydride contains less than 8 electrons.
- Electron precise hydride contains 8 electrons without lone pair.
- Electron rich hydride contains 8 electrons along with lone pair(s).

**74.** *(c)* The correct match from the given options is option (c), i.e.

A-(ii), B-(iii), C-(i), D-(iv)

Li — Electrochemical cell

Na — Coolant in fast breeder reactors

KOH — Absorbent for $CO_2$

Cs — Photoelectric cell

- Lithium is used in electrochemical cells. These cells generate electrical energy from chemical reactions.
- Coolants are used to fill the engine's cooling system, to act as a heat exchange fluid. Sodium is one of its constituent.
- KOH is a strong base and $CO_2$ is slightly acidic. It absorbs $CO_2$ readily at room temperature.
- Caesium is used in photoelectric cells.

**75.** *(a)* **Assertion (A)** is correct because in point defects of ionic solids, electrical neutrality is an essential condition.

**Reason (R)** is also correct because cation dislocates from lattice site to interstitial position in Frenkel defect.

Hence, both (A) and (R) are correct but (R) is not the correct explanation of (A).

**76.** *(b)* The incorrect statement about enzymes is option (b) because enzymes are protein in nature and comprised of amino acids that are linked together in one or more polypeptide chains. Rest all other given options are correct.

**77.** *(c)* The correct match is option (c), i.e.

(A)-(iv), (B)-(iii), (C)-(ii), (D)-(i)

| Products formed | Reaction of carbonyl compound with |
|---|---|
| Cyanohydrin | HCN |
| Acetal | Alcohol |
| Schiff's base | $RNH_2$ |
| Oxime | $NH_2OH$ |

The reactions involved are as follows

$$>C=O + HCN \xrightarrow{OH^-} >C<\begin{matrix}OH\\CN\end{matrix}$$ Cyanohydrin

$$\begin{matrix}R\\H\end{matrix}>C=O + 2ROH \xrightarrow[\text{(Alcohol)}]{OH^-} \begin{matrix}R\\H\end{matrix}>C<\begin{matrix}OR\\OR\end{matrix}$$ Acetal

$$>C=O + R-NH_2 \xrightarrow{H^+} \begin{matrix}R\\H\end{matrix}>C=N-R$$ Schiff's base

$$>C=O + NH_2OH \xrightarrow{H^+} \begin{matrix}R\\H\end{matrix}>C=N-OH$$ Oxime

**78.** *(a)* We know that,

$$\text{Molality} = \frac{\text{number of moles of solute}}{\text{weight of the solvent (g)}} \times 1000$$

Given that molality = 1

Moles of solute = 0·5

$$1 = \frac{0\cdot 5}{\text{Weight of the solvent (g)}} \times 1000$$

Weight of the solvent (g) = 500 g of solvent.

Hence, option (a) is correct.

**79.** *(b)* The correct statement is option (b). It is because in diamond, each carbon is bonded with four other carbon atoms. So, hybridisation of carbon atom is $sp^3$. In graphite each carbon atom is bonded with three other carbon atoms.

So, hybridisation of carbon atom in graphite is $sp^2$.

Hence diamond is $sp^3$-hybridised and graphite is $sp^2$-hybridised. Rest all other options are incorrect. It is because only graphite is used as dry lubricants. Diamond has 3-D network structure. Both diamond and graphite are covalent.

**80.** *(c)* Among the given options, the incorrect statement is option (c).

$O_2^+ = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2$

$\pi 2p_x^2 \pi 2p_y^2 \pi^* 2p^1$

$O_2^+$ ions have 15 electrons, so it contains one unpaired electron. Hence, it is paramagnetic in nature.

**81.** *(d)* In interhalogen compound of group 17th, ICl is more reactive than $I_2$ due to polar bonds. Also, $X—X'$ bond is weaker than $X—X$ bond except $F_2$. So I— Cl bond is weaker than I—I bond.

Hence, both A and R are correct and R is the correct explanation of A.

**82.** *(d)* Boiling point of comparable molecular mass molecules is as follows

$R$—OH > aldehyde-ketone > alkane.

The order followed above is due to presence of hydrogen bonding (strong molecular association) in alcohol. In aldehyde-ketone, dipole-dipole interactions are present (weak molecular association) while in alkane it is non-polar hence, have lowest boiling point among all three.

Hence, both statement I and statement II are correct. So, the correct option is (d).

**83.** *(d)* Given that, $pK_a$ of

$CH_3COOH = 4.57$

Concentration of acetic acid $(CH_3COOH) = 0.01$ M

Concentration of sodium acetate $(CH_3COONa) = 0.10$ M

Weak acid ($CH_3COOH$) and salt of weak acid (i.e. strong base, $CH_3COONa$) forms an acidic buffer.

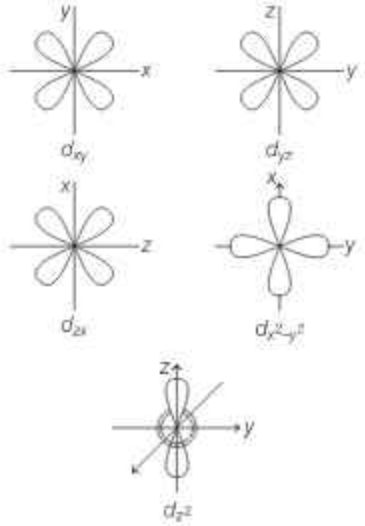
pH of acidic buffer solution is given by

$$pH = pK_a + \log\frac{[\text{Salt}]}{[\text{Acid}]}$$

$$= 4.57 + \log\left(\frac{0.1}{0.01}\right) = 4.57 + 1 = 5.57.$$

Hence, the given solution has a pH 5.57.

**84.** *(c)* The incorrect statement is option (c) because the shape of first four orbitals, i.e. $d_{xy}$, $d_{yz}$, $d_{zx}$ and $d_{x^2-y^2}$ are similar to each other, that is, double dumb bell shape, whereas the fifth one, i.e. $dz^2$ is different from other.



**85.** *(c)* Standard reduction potential values is given to us which can be arranged as follows

$E^\circ_{Zn^{2+}/Zn} < E^\circ_{Fe^{2+}/Fe} < E^\circ_{Cu^{2+}/Cu} < E^\circ_{Ag^+/Ag}$

So, the reactivity order will be Zn > Fe > Cu > Ag.

The reactions given in the options are displacement reaction. In case of displacement reaction, more reactive metals (i.e., low SRP) can displace less reactive metals (higher SRP) from their salt solution.

So, $CuSO_4(aq) + 2Ag(s) \longrightarrow Cu(s) + Ag_2SO_4(aq)$ is not possible.

As Ag is less reactive as compared to Cu.

Hence, the correct option is (c).

**86.** *(b)* en is a strong field ligand. As the number of en (strong field ligand) increases, splitting also increases. So, $\Delta_o$ increases. Hence, $[Ni(en)_3]^{2+}$ will have highest energy due to presence of 3 en ligands followed by $[Ni(H_2O)_2(en)_2]^{2+}$ and $[Ni(H_2O)_4en]^{2+}$.

So, the correct order is (C) > (A) > (B).

**87.** *(c)* Given

$V = 10\,L$

$W_{O_2} = 64\,g$

$T = 27^\circ C = 27 + 273 = 300\,K$

$R = 0.0831\,L\,bar\,K^{-1}\,mol^{-1}$

$n_{O_2} = 2$

Using, ideal gas equation

$$pV = nRT$$

$$\Rightarrow \quad p = \frac{nRT}{V}$$

Substituting the values

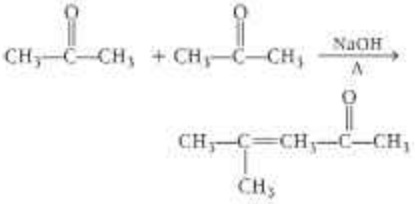
$$p = \frac{2 \times 0.0831 \times 300}{10}$$

$$p = 4.986 \approx 4.9\,bar$$

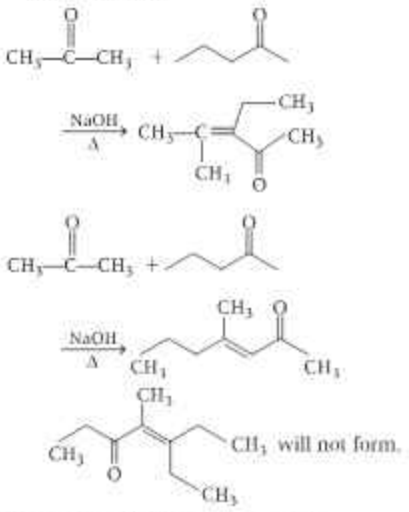
Hence option (c) is correct.

**88.** *(a)* When acetone reacts with 2-pentanone in presence of dilute NaOH followed by heating then the products will be formed according to aldol condensation.

**Self aldol**



**Cross aldol**




Hence, (a) is the correct option.

**89.** *(*)* Given, $E^\circ_{cell} = 10\text{-}5\,V$

$$\frac{2.303\,RT}{F} = 0.059$$

$[Ni^{2+}] = 0.001\,M = 10^{-3}\,M$

$[Ag^+] = 0.001\,M = 10^{-3}\,M$

Using Nernst equation

$$E_{cell} = E^\circ_{cell} - \frac{0.059}{n}\log\frac{[Ni^{+2}]^1}{[Ag^+]^2}$$

$$= 10.5 - \frac{0.059}{2}\log\frac{(10^{-3})}{(10^{-3})^2}$$

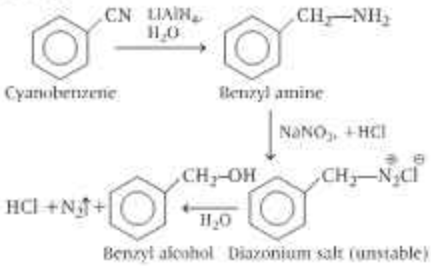
$$= 10.5 - \frac{0.059}{2}\log 10^3$$

$$= 10.5 - \frac{0.059}{2} \times 3$$

$$= 10.5 - 0.0885$$

$$E_{cell} = 10.4115\,V$$

**90.** *(c)*




In this reaction cyanobenzene reacts with strong reducing agent, i.e. $LiAlH_4$ to yield benzyl amine which on further reaction with $NaNO_2$ + HCl will give diazonium salt. This salt on reaction with $H_2O$ will yield benzyl alcohol.

Hence, the correct option is (c).

**91.** *(d)* Given

$a = 3.608 \times 10^{-8}\,cm$

$d = 8.92\,g/cm^3$

fcc unit cell,

Hence, $Z = 4$

Atomic mass = ?

Using the formula of density,

$$d = \frac{Z \times M}{N_A \times a^3}$$

where, $d$ = density

$M$ = atomic mass

$a$ = edge length

$N_A$ = Avogadro's number

$Z$ = number of atoms in unit cell

Now, substituting the values in the above formula,

$$8.92 = \frac{4 \times M}{6.022 \times 10^{23} \times (3.608 \times 10^{-8})^3}$$

$M = 63.1\,g/mol$ (Molar atomic mass)

Or Atomic mass ($M$) = 63.1 u.

Hence, option (d) is correct.

**92.** *(d)* Given, $K_C = 3 \times 10^{-59}$

$[O_2] = 0.040\,M$

$3O_2(g) \rightleftharpoons 2O_3(g)$

$$K_C = \frac{[O_3]^2}{[O_2]^3}$$

Substituting the values,

$$3 \times 10^{-59} = \frac{[O_3]^2}{(4 \times 10^{-2})^3}$$

$[O_3]^2 = 3 \times 10^{-59} \times (4 \times 10^{-2})^3$

$[O_3]^2 = 19.2 \times 10^{-64}$

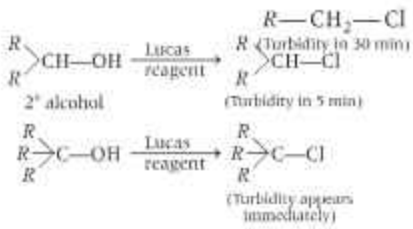
$[O_3] = 4.38 \times 10^{-32}\,M$

Hence, the correct option is (d).

**93.** *(b)* 1°, 2°, 3° alcohols can be distinguished by using Lucas reagents (conc. HCl + anhy. $ZnCl_2$) on the basis of the time taken for turbidity to appear,

i.e. $R—CH_2—OH \xrightarrow{\text{Lucas reagent}} R—CH_2—Cl$

1° alcohol (Turbidity in 30 min)




Hence the order of reactivity of alcohol towards Lucas reagent is in the order 3° > 2° > 1°.

So, statement I is correct but statement II is incorrect. Option (b) is correct.

**94.** *(d)* The reaction that takes place in the given condition is as follows

$$\underset{+7}{KMnO_4} + I^- \xrightarrow[\text{Weak alkaline medium}]{\text{Neutral}} \underset{+4}{MnO_2} + IO_3^-$$

So, the change in oxidation state of manganese in this reaction is from + 7 to + 4.

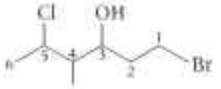
Hence, option (d) is correct option.

**95.** *(a)* The correct match is

A-(iii), B-(i), C-(ii), D-(iv)

- Haematite is an ore of iron. its composition is $Fe_2O_3$.
- Magnetite also, is an ore of Iron. Its composition is $Fe_3O_4$.
- Calamine is an ore of zinc. Its composition is $ZnCO_3$.
- Kaolinite is a clay mineral with chemical formula $[Al_2(OH)_4Si_2O_5]$.

**96.** *(d)*




The IUPAC name of the given compound is 1-bromo-5-chloro-4 methylhexan-3-ol.

**97.** *(d)* According to Bohr's atomic model

$$r \propto \frac{n^2}{Z}$$

For 2nd orbit of $He^+$ $n_2 = 2$; $z_2 = 2$

for 3rd orbit of $Li^{2+}$ $n_1 = 3$

$$z_1 = 3$$

From the above equation

$$\frac{(r_3)_{Li^{2+}}}{(r_2)_{He^+}} = \frac{n_1^2 \times z_2}{n_2^2 \times z_1}$$

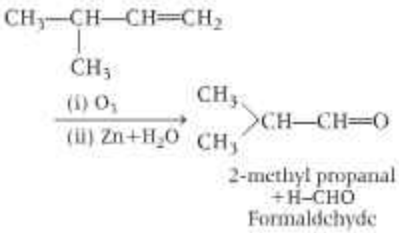
$$\frac{(r_3)_{Li^{2+}}}{105 \cdot 8 \text{ pm}} = \frac{3 \times 3 \times 2/3}{2 \times 2}$$

$$(r_3)_{Li^{2+}} = \frac{3 \times 3 \times 2 \times 105 \cdot 8}{4 \times 3}$$

$$(r_3)_{Li^{2+}} = 158 \cdot 7 \text{ pm}.$$

Hence, the radius of third Bohr orbit of $Li^{2+}$ ion is 158.7 pm.

**98.** *(d)* The compound '*X*' that reacts with $O_3$ followed by reaction with $Zn / H_2O$ gives formaldehyde and 2-methyl propanal as products is 3-methyl but-1-ene. The ozonolysis reaction that involved in this is as follows




**99.** *(a)* Given,

Initial concentration $[A_0] = 0.1$ M

Conc, after 5 min $[A_t] = 0.001$ M

$t = 5$ minutes

Using rate constant formula for 1st order reaction

$$K = \frac{2.303}{t} \log\left(\frac{A_0}{A_t}\right)$$

$$= \frac{2.303}{5} \log\left(\frac{0.1}{0.001}\right)$$

$$K = \frac{2.303 \times 2}{5} = 0.9212 \text{ min}^{-1}$$

Hence, the rate constant for the reaction is $K = 0.9212 \text{ min}^{-1}$.

**100.** *(a)* The pollution due to oxides of sulphur gets enhanced due to the presence of particulate matter, ozone and hydrogen peroxide.

The particulate matter present in polluted air catalyses the oxidation of sulphur dioxide to sulphur trioxide

$$2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$$

Similarly, the reaction is also promoted by ozone and hydrogen peroxide.

$$SO_2(g) + O_3(g) \longrightarrow SO_3(g) + O_2(g)$$

$$SO_2(g) + H_2O_2(l) \longrightarrow H_2SO_4(aq)$$

Hence, option (a) is correct.

## Botany

**101.** *(a)* National parks are not a method of *ex-situ* conservation. These are sites of protecting an endangered plant or animal species in its natural habitat. *Ex-situ* conservation literally means, "off-site conservation". It is the process of protecting an endangered species, variety or breed, of plant or animal outside of its natural habitat, For example, by removing part of the population from a threatened habitat and placing it in a new location, which may be a wild area or within the care of humans. For example, cryopreservation, micropropagation, *in-vitro* fertilisation, zoological parks, etc.

**102.** *(d)* Both statements I and II are correct. The primary $CO_2$ acceptor is a 3-carbon molecule phosphoenolpyruvate (PEP) in $C_4$ plants and is present in the mesophyll cells. The enzyme responsible for this fixation is PEP carboxylase or PEP case. Also, mesophyll cells lack RuBisCO enzyme. The $C_4$ acid is OAA, which is formed in the mesophyll cells. It then forms other 4-carbon compounds like malic acid or aspartic acid in the mesophyll cells itself, which are transported to the bundle sheath cells. In the bundle sheath cells these $C_4$ acids are broken down to release $CO_2$ and a 3-carbon molecule.

**103.** *(b)* The XO type of sex-determination can be found in grasshoppers , crickets and some other insects. In this type of insects, the females are homogametic, which means they produce only one single type of chromosome that is XX. Whereas the male have only one X chromosome.

**104.** *(d)* Options (I) and (II) are correct. In older trees the greater part of secondary xylem is dark brown due to the deposition of tannins, resins, oil, gums, aromatic substances and essential oil in the central or the innermost layer of the stem. It also has secretion of secondary metabolites and their deposition in lumen of vessels. These substances make it hard durable and resistant to the attacks of microorganisms and insects.

**105.** *(e)* The apoplast does not provide any barrier to water movement and water movement is through mass flow. The apoplastic movement of water occurs exclusively through the intercellular spaces and the walls of the cells. Movement through the apoplast does not involve crossing the cell membrane. This movement is dependent on the gradient. As water evaporates into the intercellular spaces or the atmosphere, tension develop in the continuous stream of water in the apoplast, hence mass flow of water occurs due to the adhesive and cohesive properties of water.

**106.** *(a)* Both (A) and (R) are correct, but (R) is not the correct explanation of (A). Polymerase chain reaction, is a method of amplifying the fragments of DNA. This method can make multiple copies of even a single DNA fragment or the gene of interest in a test tube. If we spread the transformed cells on agar plates containing ampicillin, only transformants will grow, untransformed recipient cells will die. Thus, due to ampicillin resistance gene, one is able to select a transformed cell in the presence of ampicillin. The ampicillin resistance gene in this case is called a selectable marker.

**107.** *(b)* Statement (b) is not related to gel electrophoresis. DNA fragments can be separated by a technique known as gel electrophoresis. Since DNA fragments are negatively charged molecules they can be separated by forcing them to move towards the anode under an electric field through a medium/matrix. The most commonly used matrix for gel electrophoresis is agarose which is a natural polymer extracted from sea weeds. The DNA fragments are separated according to their size through sieving effect provided by the agarose gel. The smaller the fragment size, the farther it moves. The separated DNA fragments can be visualised as bright orange coloured bands after staining the DNA with a compound known as ethidium bromide followed by exposure to UV radiation. The separated bands of DNA are cut out from the agarose gel and extracted from the gel piece. This step is known as elution.

**108.** *(a)* Ethylene is a simple gaseous PGR. It is synthesised in large amounts by tissues undergoing senescence and ripening fruits. It promotes root growth and root hair formation, thus helping the plants to increase their absorption surface. Ethylene breaks seed and bud dormancy, initiates germination in peanut seeds, sprouting of potato tubers and promotes rapid internode/petiole elongation in deep water rice plants. It also helps leaves/ upper parts of the shoot to remain above water.

**109.** *(c)* In lactic acid fermentation not much energy is released i.e., less than 7 per cent of the energy in glucose is released and not all of it is trapped as high energy bonds of ATP. Also, the processes are hazardous – either acid or alcohol is produced.

**110.** *(d)* All statements (I), (II), (III), (IV) and (V) are correct. When xylem and phloem within a vascular bundle are arranged in an alternate manner along the different radii, the arrangement is called radial arrangement such as in roots. Conjoint closed vascular bundles do not possess cambium and in open vascular bundles, cambium is present in between xylem and phloem. The vascular bundles of dicotyledonous stem possess endarch protoxylem. In monocotyledonous root, usually there are more than six xylem bundles present, i.e. polyarch condition.

**111.** *(b)* Statements (II), (III), (IV) and (V) are correct. Axillary buds form slender and spirally coiled tendrils in cucumber and pumpkin. Stem is flattened and fleshy in *Opuntia* and modified to perform the function of leaves. *Rhizophora* shows vertically upward growing roots that help to get oxygen for respiration. Subaerially growing stems in grasses and strawberry help in vegetative propagation. Incorrect statement (a) can be corrected as In *Citrus* and *Bougainvillea* the three terminal leaflets of leaf get modified into claw like hooks which help in climbing.

**112.** *(c)* Plants follow different pathways in response to environment or phases of life to form different kinds of structures. This ability is called plasticity. For example, in cotton, buttercup, coriander and larkspur plants, the leaves of the juvenile phase are different in shape from those in mature phase it is known as heterophylly.

**113.** *(d)* Both statement I and statement II are correct. These can be explained as cleistogamous flowers are flowers, which do not open at all. In such flowers, the anthers and stigma lie close to each other. When anthers dehisce in the flower buds, pollen grains come in contact with the stigma for effective pollination. Thus, cleistogamous flowers are invariably autogamous as there is no chance of cross-pollen landing on the stigma. It is a big disadvantage of being closed always. Cleistogamous flowers also produce assured seed-set even in the absence of pollinators.

**114.** *(a)* "Girdling Experiment" was performed by plant physiologists to identify the plant tissue through which food is transported.In this experiment, on the trunk of a tree a ring of bark up to a depth of the phloem layer is carefully removed. In the absence of downward movement of food the portion of the bark above the ring on the stem becomes swollen after a few weeks. This experiment shows that phloem is the tissue which is responsible for translocation of food and that transport takes place in one direction, i.e., towards the roots.

**115.** *(b)* Hydrocolloid carrageen is obtained from Rhodophyceae (red algae) only. Certain marine brown and red algae produce large amounts of hydrocolloids (water holding substances), e.g., algin (brown algae) and carrageen (red algae) which are used commercially.

**116.** *(d)* Both statement I and statement II are correct. Gregor Johann Mendel, conducted hybridisation experiments on garden peas for seven years (1856-1863) and proposed the laws of inheritance in living organisms. For his experiments, he studied seven pairs of contrasting traits in pea plants. These were seed shape and colour, flower colour, pod shape and colour, flower position and stem height.

**117.** *(b)* Ethylene is a simple gaseous PGR. It is synthesised in large amounts by tissues undergoing senescence and ripening fruits. It promotes female flowers in cucumbers thereby increasing the yield.

**118.** *(d)* The process of translation of *m*RNA to proteins begins as soon as the small subunit of ribosome encounters *m*RNA.Translation refers to the process of polymerisation of amino acids to form a polypeptide.The cellular factory responsible for synthesising proteins is the ribosome. The ribosome consists of structural RNAs and about 80 different proteins. In its inactive state, it exists as two subunits, a large subunit and a small subunit. When the small subunit encounters an *m*RNA, the process of translation of the *m*RNA to protein begins.

**119.** *(b)* Habitat loss and fragmentation, over exploitation, alien-species invasion and co-extinction are causes for biodiversity loss. These are "Evil Quartet" of biodiversity loss. In general, loss of biodiversity in a region may lead to decline in plant production, lowered resistance to environmental perturbations such as drought and increased variability in certain ecosystem processes such as plant productivity, water use, and pest and disease cycles.

**120.** *(a)* The set of correct statements are (I), (III) and (IV). The incorrect statements can be corrected as The heterochromatin is more densely packed and stains dark. Euchromatin is said to be transcriptionally active chromatin, whereas heterochromatin is inactive. A typical nucleosome contains 200 bp of DNA helix.

**121.** *(a)* Option (a) is incorrectly matched, it can be corrected as *Ulothrix* belongs to class–Chlorophyceae and the stored food material in them is starch. Mannitol is present in members of class–Phaeophyceae.

**122.** *(b)* The device which can remove particulate matter present in the exhaust from a thermal power plant is electrostatic precipitator. It can remove over 99 per cent particulate matter present in the exhaust from a thermal power plant. It has electrode wires that are maintained at several thousand volts, which produce a corona that releases electrons. These electrons attach to dust particles giving them a net negative charge. The collecting plates are grounded and attract the charged dust particles.

**123.** *(b)* DNA polymorphism forms the basis of both genetic mapping and DNA fingerprinting. DNA fingerprinting is a technique to find out variation in individuals of a population at DNA level. It works on the principle of polymorphism in DNA sequences. It has immense applications in the field of forensic science, genetic biodiversity and evolutionary biology. Gene mapping refers to the process of determining the location of genes on chromosomes. It involves sequencing a genome and then using computer programs to analyse the sequence to identify the location of genes.

**124.** *(b)* Statement (b) cannot be connected to predation.In predation only one species is negatively impacted and other is completely benefitted. Predators also help in maintaining species diversity in a community, by reducing the intensity of competition among competing prey species. Predation might lead to extinction of a species. But it is necessitated by nature to maintain the ecological balance.

**125.** *(b)* Pairing of homologous chromosomes never occurs during mitotic cell division.Mitosis or the equational division is usually restricted to the diploid cells only. In this, spindle fibres attach to kinetochores of chromosomes, movement of centrioles towards opposite poles occurs and coiling and condensation of the chromatids occurs.

**126.** *(b)* Exoskeletons of arthropods is composed of chitin. Arthropods are the most successful group of animals. Their success is due to unique chitinous cuticle. Exoskeleton is light weight , tough and composed of structural polysaccharide chitin . It is strengthened with proteins and calcium carbonate occurring on the outside.

**127.** *(b)* The net gain of ATP when each molecule of glucose is converted to two molecules of pyruvic acid is 2 ATP.

**128.** *(a)* Vexillary aestivation and diadelphous dithecous anther is seen in *Pisum sativum*. In them, there are five petals, the largest (standard) overlaps the two lateral petals (wings) which in turn, overlap the two smallest anterior petals (keel), this type of aestivation is known as vexillary. When the stamens are united into two bundles, it is diadelphous.

**129.** *(c)* Statement (c) is incorrect, it can be corrected as honeybees are the most dominant pollinating agents among animals. Rest statements are correct.

**130.** *(b)* Statement I is correct, but statement II is incorrect. Incorrect statement can be corrected as decomposition is largely an oxygen-requiring process. The rate of decomposition is controlled by chemical composition of detritus and climatic factors. In a particular climatic condition, decomposition rate is slower, if detritus is rich in lignin and chitin, and is quicker, if detritus is rich in nitrogen and water-soluble substances like sugars.

**131.** *(b)* The appearance of recombination nodules on homologous chromosomes during meiosis characterises sites at which crossing over occurs between non-sister chromatids of the homologous chromosomes. Crossing over is the exchange of genetic material between two homologous chromosomes.

**132.** *(a) Frankia* produces nitrogen-fixing nodules on the roots of non-leguminous plants like *Alnus*.

**133.** *(a)* Statement (a) is not related to chemiosmosis. In chemiosmosis there is breakdown of proton gradient not electron gradient. Proton gradient is important because it is the breakdown of this gradient that leads to the synthesis of ATP. The gradient is broken down due to the movement of protons across the membrane to the stroma through the transmembrane channel of the $CF_0$ of the ATP synthase. The NADP reductase enzyme is located on the stroma side of the membrane.

**134.** *(a)* The flowers are zygomorphic in gulmohar and *Cassia*. Zygomorphic flowers are those flowers that have their floral parts unequal in size or form so that the flower is capable of division into essentially symmetrical halves by only one longitudinal plane passing through the axis.

**135.** *(a)* The correct match are
(A) – (iv), (B) – (iii), (C) – (ii), (D) – (i).
Manganese-Functions in splitting of water during photosynthesis,
Magnesium-Activates enzymes of respiration
Boron-Required for pollen germination,
Iron-Activates the enzyme catalase.

**136.** *(a)* Addition of more solutes in a given solution will lower its water potential. Water potential is the chemical potential of water and depends on the number of water molecules. When a solute has been dissolved in water, there is a relative decrease in number of solvent or water molecules. Hence, water potential

decreases. The water potential of pure water has been fixed as zero. The water potential of all solutions is negative.

**137.** *(b)* Statements III, IV and V are correct. Incorrect statements can be corrected as, some lipids have phosphorus and a phosphorylated organic compound in them. These are phospholipids. They are found in cell membrane like lecithin. Fatty acids could be saturated (without double bond) or unsaturated (with one or more C=C double bonds).

**138.** *(a)* The role of large bundle sheath cells found around the vascular bundles in $C_4$ plants is to increase the number of chloroplast for the operation of Calvin cycle. The leaves which have such anatomy are said to have 'Kranz' anatomy. 'Kranz' means 'wreath' and is a reflection of the arrangement of cells. The bundle sheath cells may form several layers around the vascular bundles; they are characterised by having a large number of chloroplasts, thick walls impervious to gaseous exchange and no intercellular spaces.

**139.** *(a)* Statement (a) is false. All the buses of Delhi were converted to run on CNG from diesel by the end of 2002. This was done because CNG is better than diesel. CNG burns most efficiently, unlike petrol or diesel, in the automobiles and very little of it is left unburnt.

Moreover, CNG is cheaper than petrol or diesel, cannot be siphoned off by thieves and adulterated like petrol or diesel. The main problem with switching over to CNG is the difficulty of laying down pipelines to deliver CNG through distribution points/ pumps and ensuring uninterrupted supply.

**140.** *(a)* Transposons can be used during gene silencing.Transposons or mobile genetic elements in viruses are the sources of the complementary *ds*RNA, which in turn bind to specific *m*RNA and cause RNA interference (RNA*i*). RNA*i* takes place in all eukaryotic organisms as a method of cellular defence.

**141.** *(d)* The correct match are (A) – (iii), (B) – (i), (C) – (iv), (D) – (ii)

- Metacentric chromosome – Centromere in the middle forming two equal arms of chromosomes.
- Acrocentric chromosome – Centromere situated close to the end forming one extremely short and one very long arms.
- Sub-metacentric – Centromere slightly away from the middle forming one shorter arm and one longer arm.
- Telocentric chromosome – Centromere at the terminal end.

**142.** *(a)* The correct match are (A) – (ii), (B) – (iii), (C) – (iv), (D) – (i)

*Spirogyra* – Dominant haploid free-living gametophyte.

Fern – Dominant diploid sporophyte alternating with reduced gametophyte called prothallus.

*Funaria* – Dominant haploid leafy gametophyte alternating with partially dependent multicellular sporophyte.

*Cycas* – Dominant diploid sporophyte vascular plant, with highly reduced male or female gametophyte.

**143.** *(a)* Myotonic dystrophy occurs due to the presence of autosome linked dominant trait, this means one copy of the altered gene in each cell is sufficient to cause the disorder. In most cases, an affected person has one parent with the condition.

**144.** *(a)* Statements I, III, IV and V are related to spring wood. These statements can be explained as in spring season, cambium is very active and produces a large number of xylary elements having vessels with wider cavities. The wood formed during this season is called spring wood or earlywood. The spring wood is lighter in colour and has a lower density, whereas the autumn wood is darker and has a higher density.

**145.** *(b)* The part labelled as C (thalamus) in the figure makes it a false fruit.A false fruit is derived from the floral parts other than ovary like thalamus. False fruit is also called pseudocarp or accessory fruit or spurious fruit.

**146.** *(a)* 5′ G A A T T C 3′ 3′ C T T A A G 5′

The above given sequence is a palindromic sequences of DNA, which can be cut easily by particular restriction enzyme. The palindrome in DNA is a sequence of base pairs that reads same on the two strands when orientation of reading is kept the same.

**147.** *(d)* Predation is an interaction in which we can assigned (+) for one species and (-) for another species involved. Other options can explained as

Both the species are in loss in competition. The interaction where one species is benefitted and the other is neither benefitted nor harmed is called commensalism. In amensalism, on the other hand one species is harmed, whereas the other is unaffected.

**148.** *(b)* Weathering of rocks will accelerate phosphorus cycle. The natural reservoir of phosphorus is rock, which contains phosphorus in the form of phosphates. When rocks are weathered, minute amounts of these phosphates dissolve in soil solution and are absorbed by the roots of the plants. Phosphorus is a major constituent of biological membranes, nucleic acids and cellular energy transfer systems. Many animals also need large quantities of this element to make shells, bones and teeth.

**149.** *(b)* (A) is correct, but (R) is not correct. (R) can be explained as Mendel's law of independent assortment does not hold good for the genes that are located closely on the same chromosome. These genes are called as 'linked genes'. Closely located genes assorted together, and distantly located genes, due to recombination, assorted independently. Linkage maps, therefore, corresponded to arrangement of genes on a chromosome.

**150.** *(d)* Sequence annotation will be used by geneticist as a blind approach for sequencing the whole set of genome that contained all the coding and non-coding sequence, and later assigning different regions in the sequence with functions. For sequencing, the total DNA from a cell is isolated and converted into random fragments of relatively smaller sizes and cloned in suitable host using specialised vectors. The cloning resulted into amplification of each piece of DNA fragment so that it subsequently could be sequenced with easily.

## Zoology

**151.** *(b)* Statement I is correct, but statement II is incorrect.

Autoimmune disorders are abnormal immune responses in which the immune system fails to properly differentiate between self and non-self cells and destroys body's own protein.

Rheumatoid arthritis is an autoimmune disorder that primarily affects joints. Thus, joints becomes painful, stiff and swollen.

**152.** *(c)* Statement I is incorrect, but statement II is correct.

A blood clot or coagulum is formed mainly of a network of threads called fibrins in which dead and damaged formed elements of blood are trapped.

Spleen is the graveyard of RBCs because after completion of lifespan, old and damaged RBCs or erythrocytes are destroyed in spleen where they are ingested by free macrophages.

**153.** *(a)* Conidia is the asexual reproductive structure associated with *Penicillium*.

Asexual reproduction refers to the production of offspring by a single parent without the fusion of gametes. Conidia are non-motile gametes that are produced exogenously.

**154.** *(b)* Statement I is correct, but statement II is incorrect.

Spermiation is the process of releasing mature spermatozoa from the Sertoli cells into the lumen of seminiferous tubules. Spermiogenesis is the process of transformation of spermatids to spermatozoa or sperms.

**155.** *(a)* About 20 mL of oxygen is present in 100 mL of blood, when it reaches to tissues only 5 mL of oxygen is delivered and 15 mL returns back.

Thus, under normal physiological conditions in human being every 100 mL of oxygenated blood can deliver 5 mL of $O_2$ to the tissues.

**156.** *(d) In-situ* conservation involves in the protection

of species in their natural habitat. It is the conservation and protection of whole ecosystem and its biodiversity at all levels in order to protect the threatened species. For example, National parks, Wildlife sanctuaries, etc.

**157.** *(a)* Natural selection where more individuals acquire specific character value other than the mean character value, leads to directional change. This operates in response to gradual changes in environmental condition like during industrial revolution in peppered moths.

**158.** *(b)* Biofortification is the method of breeding crops with higher level of nutrients to improve public health. The objective of breeding for improved nutritional quality is to enhance protein and oil content, vitamin content and micronutrient and mineral content. For example, Atlas-66 is a protein rich wheat variety.

**159.** *(a)* Cartilage is a flexible connective tissue present between the adjacent bones of the vertebral column. It provides cushioning effect and reduces friction between the bones during movement.

**160.** *(c) Pavo* belongs to class-Aves and thus, excretes nitrogenous waste as uric acid in the form of pellet or paste with minimum loss of water.

**161.** *(b)* Prokaryotes are organisms that do not have a nuclear membrane and other membrane bound organelles. Thus, prokaryotes do not have RER and SER.

**162.** *(c)* Statements (II), (III) and (V) are incorrect.

Oogenesis is the process of formation of mature female gamete. Differentiation of gamete will not occur after the completion of meiosis. Meiosis occurs continuously in a mitotically dividing egg mother cells or oogonia. It is controlled by the gonadotropin (LH and FSH) secreted by the anterior lobe of pituitary gland. It is initiated even before birth, i.e., during the embryonic development stage, but is completed only after fertilisation.

**163.** *(d)* Both (A) and (R) are correct and (R) is the correct explanation of (A).

The characteristic feature of phylum–Chordata includes the presence of a notochord and paired pharyngeal gill slits. In sub-phylum–Vertebrata, notochord present in embryo gets replaced by a cartilaginous or bony vertebral column in adults. But in cephalochordates, notochord persists throughout life as such and in urochordates, notochord is present only in larval stages and absent in adults. Thus, it can be said that all vertebrates are chordates but all chordates are not vertebrates.

**164.** *(c)* Digestive tract of animals like *Pavo, Psittacula* and *Corvus* has

additional chambers like crop and gizzard. These animals belong to class–Aves and animals of this class (birds) store their food in crop after swallowing. Then food is passed to the true stomach or proventricules. The food then gets passed to the gizzard and then back to proventricules. The function of gizzard is to chew and grind the food.

**165.** *(b)* (A) is correct, but (R) is not correct.

Osteoporosis is characterised by decreased bone mass and increased chances of fractures. It results in weakness of skeletal muscles. It is caused due to excessive reabsorption of calcium and phosphorus from bone. The decreased level of oestrogen is common cause of this.

**166.** *(c)* ADA (Adenosine deaminase) deficiency is a rare inherited disorder of purine metabolism characterised by the accumulation of metabolic substrates that lead to hampering of the functioning of immune system and causes Severe Combined Immunodeficiency (SCID).

Gene therapy is the replacement of defective or faulty gene with a normal or healthy gene to correct a genetic disorder.

In gene therapy of ADA deficiency, the patient requires periodic infusion of genetically engineered lymphocytes because these cells are not immortal.

**167.** *(a)* Statement (a) is incorrect with reference to mitosis because spindle fibres attach to kinetochores of chromosomes in metaphase.

**168.** *(d)* The hierarchical arrangement given in option (4) is correct in case of animals. Thus, the correct arrangement in ascending order is

Kingdom → Phylum → Class → Order → Family → Genus → Species.

**169.** *(d)* Both statement I and statement II are correct.

Each restriction endonuclease functions by 'inspecting' the length of a DNA sequence. Once it finds specific recognition sequence, it will bind to

DNA and cut each of the two strands of double helix at specific point in their sugar phosphate backbone. Thus, each restriction endonuclease recognises specific sequence to cut DNA known as palindromic sequence.

Restriction enzymes cut the strand of DNA a little away from centre of the palindrome sites, but between the same two bases on opposite strands.

**170.** *(b)* Statement I is correct but statement II is incorrect.

*Mycoplasma* are organisms that completely lack a cell wall. These organisms are often called PPLOs (Pleuropneumonia like Organisms). These are the smallest free-living microorganisms (0.1 – 0.5 μm) and are so small that they can easily pass through bacteria-proof filters.

**171.** *(c)* Neuroglia cells are part of neural tissue. These are the supporting cells which form a packing around the neurons in the brain, spinal cord and ganglia. Neuroglia makes up more than one half of the volume of neural tissue in our body.

**172.** *(b)* Lippes loop is a type of contraceptive used as non-medicated IUD. IUDs (Intra Uterine Devices) contain either copper or progesterone and are inserted by doctors in the uterus through vagina.

**173.** *(a)* Oogenesis is the process of formation of a mature female gamete. The production of eggs in female begins before birth that is, during the embryonic development stage, but is completed only after fertilisation.

**174.** *(d)* Cyclosporin-A is a bioactive molecule that is produced by microorganism *Trichoderma polysporum* (fungus). Cyclosporin-A is used as an immunosuppres sive agent in organ transplant patients.

**175.** *(d)* Both statement I and statement II are correct.

Fatty acids and glycerol, being insoluble in water cannot be absorbed into the blood directly. They are first modified into small droplets called micelles which move into intestinal mucosal cells. Lacteal is a lymphatic vessel of small intestine which absorbs digested fat. Lacteals carry chylomicrons into lymphatic vessels and ultimately into the blood.

**176.** *(b)* The *i* gene synthesises the repressor of the operon. Repressor binds to the operator region of operon, preventing RNA polymerase from transcribing the operon and is inactivated by interaction with the inducer such as lactose and allolactose.

Now, if *i* gene gets mutated and its product cannot bind the inducer molecule, then the repressor will remain activated, thus blocking transcription. The repressor protein binds to operator region of operon and as a result prevents RNA polymerase from binding to operon. Hence, *z*, *y* and *a* genes will not be translated.

**177.** *(a)* The first pair of wings arise from the mesothorax and the second pair arises from metathorax. First pair of wings are called tegmina which are opaque, dark and leathery. Tegmina encloses hindwings that are transparent, membranous and are used in flight.

**178.** *(c)* The conducting part of respiratory system is responsible for providing passage for air. It conditions the incoming air by warming, moistening and cleaning it. It consists of nasopharynx, larynx, trachea, bronchi, bronchioles and terminal bronchioles. Respiratory portion on the other hand is responsible for providing surface for diffusion of $O_2$ and $CO_2$.

**179.** *(d)* Arthritis is a disease causing painful inflammation and stiffness of the joints. Other three match pairs can be corrected as :

Tetany–Low $Ca^{2+}$ level causing rapid spasms.

Myasthenia gravis–Autoimmune disorder resulting in weakening and paralysis of skeletal muscle.

Muscular dystrophy–A genetic disorder causing progressive degeneration of skeletal muscle.

**180.** *(a)* Meiosis is a type of cell division that produces haploid gametes from diploid cells. DNA replication occurs in S-phase of interphase before meiosis–I begins to produce identical sister chromatids.

**181.** *(a)* Fragmentation is a process that causes breakdown of detritus into smaller particles. It is an initial stage of decomposition. In this process, detritivores feeds on the dead organic matter, turning it into smaller fragments.

**182.** *(d)* Given that,

Length of DNA molecule = 1.1 metres

As we know,

Length of DNA = 2.2 metres

Number of base pair in a human genome (diploid cell) = $6.6 \times 10^9$ bp

Thus, 2.2 m of DNA = $6.6 \times 10^9$ bp

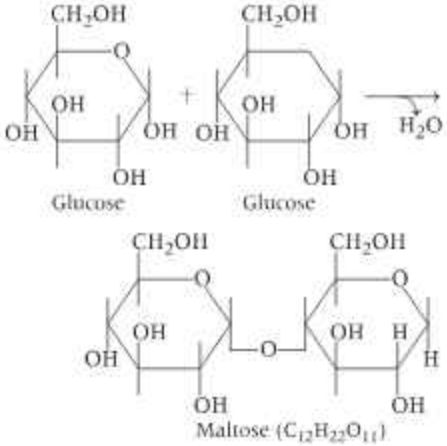
Hence, 1.1 m of DNA

$$= \frac{6.6 \times 10^9}{2} = 3.3 \times 10^9 \text{ bp}$$

**183.** *(c)* The saliva secreted into oral cavity contains electrolytes and two enzymes (salivary amylase and lysozyme). Starch is hydrolysed in oral cavity by the action of salivary amylase into a dissaccharide (i.e., maltose). Lysozyme acts as an antibacterial agent and kills bacteria thus, preventing infection.

**184.** *(b)* In dehydration reaction, when two molecules are linked together, one molecule of water ($H_2O$) is removed.

The reaction will be




**185.** *(d)* Number of *Drosophila* in laboratory = 80

Number of *Drosophila* died in a week = 08

Death rate

$$= \frac{\text{Number of individuals died}}{\text{Total number of individuals}}$$

$$= \frac{8}{80} = 0.1$$

The death rate in the population is 0.1 individuals per *Drosophila* per week.

**186.** *(c)* Statement I is incorrect, but statement II is correct. Incorrect statement can be corrected as

In a scrubber the exhaust from the thermal plant is passed through the spray of lime or water. Water dissolves the gases and lime will react with $SO_2$ to form a precipitate of calcium sulphate and sulphide.

**187.** *(d)* Cyanobacteria or blue-green algae are Gram positive autotrophic prokaryotes which perform oxygenic photosynthesis. Cyanobacteria are classified under kingdom-Monera.

Other three statements can be corrected as

- Bacteria are not exclusively heterotrophic, they can be photosynthetic or chemosynthetic autotrophs.
- Slime moulds are saprophytic organisms classified under kingdom–Protista.
- *Mycoplasma* have DNA, ribosomes but lack cell wall.

**188.** *(b)* Each division of *E.coli* takes 20 minutes.

Every 20 minutes 2 strands of DNA gets separated and a new DNA strand is synthesised using $^{14}N$ medium.

After 40 minutes, we have 2 hybrid density DNA and 2 light density DNA. After another 20 minutes i.e., 60 minutes, each of these four DNA will synthesise new DNA strand. Hence, after 60 minutes we have 2/8 DNA of hybrid density ($^{15}N - {}^{14}N$) and rest 6/8 DNA of light density ($^{14}N - {}^{14}N$).

Thus, 1 *E.coli* will form 6 *E.coli* cells that are totally free from $^{15}N$.

Thus, 10 *E.coli* will produce $10 \times 6 = 60$ *E.coli* cells free from $^{15}N$ DNA.

**189.** *(c)* Statement given in the option (c) is incorrect and can be corrected as Innate immunity is non-specific type of defence mechanism present at the time of birth.

Acquired immunity is also called specific immunity. It is a type of immunity that develops when a person's immune system responds to foreign substance or microbe, or that occurs after a person receives antibodies from another source.

**190.** *(a)* Statement given in option (II) is most appropriately correct as insulin consists of two short polypeptide chains, i.e., chain 'A' and 'B' that are linked together by disulphide bridge.

**191.** *(a)* Parathyroid hormones increases serum calcium by increasing bone reabsorption of calcium into blood. It increases renal reabsorption of calcium. Thus, PTH helps to prevent low calcium levels by acting on bones, intestine and kidneys.

**192.** *(c)* Statement (c) is incorrect and can be corrected as impulse transmission across an electrical synapse is always faster than that across a chemical synapse.

**193.** *(d)* (A)-(iv), (B)-(iii), (C)-(i), (D)-(ii)

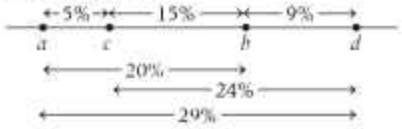
If the columnar or cuboidal cells bear cilia on their free surface they are called **ciliated epithelium**. They are mainly present in inner surface of hollow organs like **bronchioles** and Fallopian tubes.

Columnar or cuboidal cells get specialised for secretion and are called **glandular epithelium**. They are mainly of two types : unicellular consisting of isolated glandular cells (goblet cells) and multicellular consisting of isolated glandular cells (salivary gland).

Tendons which attach skeletal muscles to bones is an example of **dense regular connective tissue**.

**Adipose tissue** is a type of **loose connective tissue** located beneath the skin.

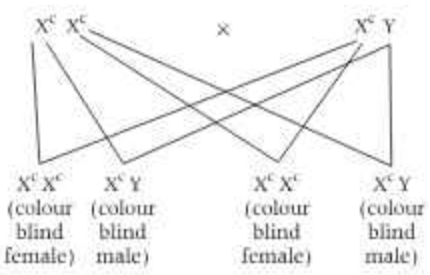
**194.** *(c)* According to the given question the sequence of genes on linear chromosome are




**195.** *(c)* Colour blindness is a X-linked recessive condition. This disease mainly affects male offspring. For a female to be born colour blind, the mother has to be atleast a carrier and the father has to be colourblind. Colourblind female is denoted by genotype $\rightarrow X^cX^c$

Man whose mother was colour blind will also be colourblind - $X^cY$

Thus, $X^cX^c \times X^cY$




**196.** *(c)* (A)-(iv), (B)-(iii), (C)-(i), (D)-(ii)

Glycogen is a polysaccharide that serves as a form of stored energy in animals and fungi.

Antibodies are immunoglobulins or gamma globulins that are large, *Y*-shaped proteins that are produced in response to foreign cells or antigens by plasma cells.

Steroids are hormones that are produced by adrenal cortex and testes. These are important for proper body functioning.

Enzymes are bioacatalytic proteins that catalyse biological reactions and are indispensable for metabolism of living beings.

**197.** *(c)* There are three important characteristics of cloning vectors

(a) Origin of replication.

(b) Selectable markers.

(c) One or more unique restriction site into

which DNA fragment can be inserted.

**198.** *(c)* Statement in option (c) is incorrect and can be corrected as flippers of penguins and dolphins are a pair of analogous organs. They have similar function, but different origin.

**199.** *(b)* Statement (b) is correct and rest other can be corrected as

As the blood fills the chamber, the atria contract to push the blood through the valve, and this is called atrial systole. It is the stage when both bicuspid and tricuspid valve opens. Increased ventricular pressure causes opening of semilunar valves. The sino-atrial node generates an action potential to stimulate atrial contraction.

**200.** *(a)* (A)-(iv), (B)-(i), (C)-(ii), (D)-(iii)

Diaphragms are made up of rubber and are inserted into female reproductive tract to cover cervix during intercourse. They prevent conception by blocking entry of sperms through cervix. Contraceptive pills alter or inhibit ovulation and implantation. They also modify the quality of cervical mucus to prevent or retard the entry of sperms.

IUDs increases phagocytosis of sperms within the uterus and can also suppress the sperm motility.

Lactational amenorrhea is the absence of menstruation during the period of intense lactation following parturition.